

Test Plan & Report Document For Timetable Management System

Testing and Validation Report for Major Project

SUBMITTED IN PARTIAL FULFILMENT FOR DEGREE OF

MASTER OF COMPUTER APPLICATIONS

By

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Test Report Cum Planning Document

Timetable Management System (TTM)



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1 Introduction

1.1 Project Overview

Project Name: Timetable Management System (TTM)

1.2 Purpose of Testing:

To ensure the system works as expected.

Identify bugs and issues in the database, backend, and frontend.

Verify data accuracy and correct API responses.

Project Status: 90% Complete (Modules are still in development)

1.3 Technologies Used

Frontend: React Vite

Backend: .NET (ASP.NET Core API)

Database: Microsoft SQL Server (MSSQL)

1.4 Testing Scope

Database Testing (Stored Procedures, Constraints, Data Validation)

API Testing (Using Postman – Verify Backend Endpoints)

UI Testing (Check Forms, Navigation, and User Interface)

Functional Testing (Ensure Timetable and Role Management Work Properly)

1.5 Testing Tools

Postman – API Testing

Microsoft SQL Server Management Studio (SSMS) – Database Testing

Google Docs – Documentation

2 Database Testing

2.1 Database Structure Validation

Test Case ID	Table Name	Test Description	Expected Output	Actual Output	Status
DB-01	Users	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-02	Faculty	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-03	Department	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-04	Subject	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-05	Timetable	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-06	WeekDay	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-07	TimeSlot	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-08	Role	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed

DB-09	RolePermission	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-10	Permission	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-11	Batch	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-12	Division	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-13	Location	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-14	SubjectType	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-15	AcademicClasses	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-16	AcademicYear	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-17	Program	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed
DB-18	StudentElectiveSubject	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	 Exists	Passed

DB-1 9	UserRole	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed
DB-2 0	StaffDetails	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed
DB-2 1	StudentDetails	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed
DB-2 2	ElectiveSubject	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed
DB-2 3	SubjectCategory	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed
DB-2 4	SubjectTeacher	Verify table existence and structure (columns, data types, primary key)	Table & correct columns exist	✓ Exists	Passed

2.2 Stored Procedures Testing

Test Case ID	Stored Procedure Name	Test Description	Expected Output	Actual Output	Status
SP-01	GetAcademicClasses	Fetch all academic classes by ProgramId	List of classes	✓ Correct	Passed
SP-02	GetCurrentAcademicYear	Fetch current academic year	Academic year data	✓ Correct	Passed
SP-03	GetBatches	Fetch all batches by DivisionId	List of batches	✓ Correct	Passed
SP-04	GetDivisions	Fetch all divisions by AcademicClassId	List of divisions	✓ Correct	Passed
SP-05	GetPrograms	Fetch all programs by DepartmentId	List of programs	✓ Correct	Passed
SP-06	GetSubjects	Fetch all subjects	List of subjects	✓ Correct	Passed
SP-07	GetLocations	Fetch all locations by DepartmentId	List of locations	✓ Correct	Passed
SP-08	GetTeachers	Fetch all teachers	List of teachers	✓ Correct	Passed

SP-09	GetDays	Fetch all available days	List of days	✓ Correct	Passed
SP-10	GetTimeSlots	Fetch all available time slots	List of time slots	✓ Correct	Passed
SP-11	GetFaculties	Fetch all faculties	List of faculties	✓ Correct	Passed
SP-12	GetDepartments	Fetch all departments	List of department s	✓ Correct	Passed
SP-13	GetLocationDetails	Fetch details of locations	List of location details	✓ Correct	Passed
SP-14	GetTimetableEntries	Fetch timetable entries	List of timetable records	✓ Correct	Passed
SP-15	GetTimetableEntry	Fetch a single timetable entry by ID	Timetable entry details	✓ Correct	Passed
SP-16	AssignSubjectToTeacher	Assign a subject to a teacher	Success message	✓ Works	Passed
SP-17	CreateRole	Insert a new role	Success message + Role added	✓ Works	Passed
SP-18	DeleteRole	Delete an existing role	Success message	✓ Works	Passed
SP-19	CreateUser	Insert a new user	Success message + User added	✓ Works	Passed

SP-2 0	DeleteUser	Delete a user by ID	Success message	✓ Works	Passed
SP-2 1	UpdateUser	Update user details	Success message	✓ Works	Passed
SP-2 2	AssignRole	Assign a role to a user	Success message	✓ Works	Passed
SP-2 3	DeassignRole	Remove a role from a user	Success message	✓ Works	Passed
SP-2 4	CreateStaffDetails	Insert new staff details	Success message	✓ Works	Passed
SP-2 5	DeleteStaffDetails	Delete staff details	Success message	✓ Works	Passed
SP-2 6	UpdateStaffDetails	Update staff details	Success message	✓ Works	Passed
SP-2 7	CreateStudentDetails	Insert new student details	Success message	✓ Works	Passed
SP-2 8	DeleteStudentDetails	Delete student details	Success message	✓ Works	Passed
SP-2 9	UpdateStudentDetails	Update student details	Success message	✓ Works	Passed
SP-3 0	AuthenticateUser	Authenticate user login credentials	Success message	✓ Works	Passed
SP-3 1	GetRoles	Fetch all roles in the system	List of roles	✓ Correct	Passed
SP-3 2	InsertTimetableEntry	Insert a new timetable entry	Success message + New entry in DB	✓ Works	Passed
SP-3 3	UpdateTimetableEntry	Modify an existing timetable entry	Success message	✓ Works	Passed

SP-3 4	DeleteTimetableEntry	Delete a timetable entry	Success message	✓ Works	Passed
SP-3 5	CheckTimeSlotConflict	Check for conflicts in timetable slots	Conflict check result	✓ Correct	Passed
SP-3 6	CheckStaffConflict	Check if a staff member is already assigned	Conflict check result	✓ Correct	Passed
SP-3 7	CheckLocationConflict	Check if a location is already booked	Conflict check result	✓ Correct	Passed
SP-3 8	CreateRole_v2.0	Insert a new role (version 2.0)	Success message + Role added	✓ Works	Passed
SP-3 9	CreateRole_v3.0	Insert a new role (version 3.0)	Success message + Role added	✓ Works	Passed
SP-4 0	DeleteRole_v2.0	Delete a role (version 2.0)	Success message	✓ Works	Passed
SP-4 1	GetPermissions	Fetch all permissions	List of permissions	✓ Correct	Passed
SP-4 2	GetRolePermissions	Fetch all permissions by role	List of role permissions	✓ Correct	Passed
SP-4 3	GetRolePermissionsByRoleId	Fetch permissions using RoleId	List of permissions for given role	✓ Correct	Passed

SP-4 4	GetRolesPermissions	Fetch roles with their permissions	List of roles and permissions	✓ Correct	Passed
SP-4 5	GetUserRoles	Fetch all roles of a user	List of user roles	✓ Correct	Passed
SP-4 6	GetUsers	Fetch all users	List of users	✓ Correct	Passed
SP-4 7	GetUsers_v2.0	Fetch all users (v2.0)	List of users	✓ Correct	Passed
SP-4 8	ToggleRoleStatus	Toggle active/inactive status of role	Success message	✓ Works	Passed
SP-4 9	ToggleUserStatus	Toggle active/inactive status of user	Success message	✓ Works	Passed
SP-5 0	UpdateRole	Update role details	Success message	✓ Works	Passed
SP-5 1	AuthenticateUser_v2.0	Authenticate user login (v2.0)	Success message	✓ Works	Passed
SP-5 2	AuthenticateUser_v3.0	Authenticate user login (v3.0)	Success message	✓ Works	Passed

SP-5 3	GetUserIdentityData_v1.0	Fetch user identity data	User identity details	✓ Correct	Passed
SP-5 4	GetUserRoleIds	Fetch role IDs for a user	List of RoleIds	✓ Correct	Passed
SP-5 5	GetStaffDetails	Fetch all staff details	List of staff records	✓ Correct	Passed
SP-5 6	GetFaculty	Fetch all faculty records	List of faculties	✓ Correct	Passed
SP-5 7	CreateFaculty	Insert new faculty	Success message + Faculty added	✓ Works	Passed
SP-5 8	UpdateFaculty	Update faculty details	Success message	✓ Works	Passed
SP-5 9	DeleteFaculty	Delete faculty by ID	Success message	✓ Works	Passed
SP-6 0	GetDepartment	Fetch all departments	List of departments	✓ Correct	Passed
SP-6 1	CreateDepartment	Insert new department	Success message + Dept added	✓ Works	Passed

SP-6 2	UpdateDepartment	Update department details	Success message	✓ Works	Passed
SP-6 3	DeleteDepartment	Delete department by ID	Success message	✓ Works	Passed
SP-6 4	GetLocation	Fetch all locations	List of locations	✓ Correct	Passed
SP-6 5	CreateLocation	Insert new location	Success message + Location added	✓ Works	Passed
SP-6 6	UpdateLocation	Update location details	Success message	✓ Works	Passed
SP-6 7	DeleteLocation	Delete location by ID	Success message	✓ Works	Passed
SP-6 8	GetProgram	Fetch all programs	List of programs	✓ Correct	Passed
SP-6 9	CreateProgram	Insert new program	Success message + Program added	✓ Works	Passed
SP-7 0	UpdateProgram	Update program details	Success message	✓ Works	Passed

SP-7 1	DeleteProgram	Delete program by ID	Success message	✓ Works	Passed
SP-7 2	GetAcademicClass	Fetch all academic classes	List of academic classes	✓ Correct	Passed
SP-7 3	CreateAcademicClass	Insert new academic class	Success message + Class added	✓ Works	Passed
SP-7 4	UpdateAcademicClass	Update academic class	Success message	✓ Works	Passed
SP-7 5	DeleteAcademicClass	Delete academic class	Success message	✓ Works	Passed
SP-7 6	GetDivision	Fetch all divisions	List of divisions	✓ Correct	Passed
SP-7 7	CreateDivision	Insert new division	Success message + Division added	✓ Works	Passed
SP-7 8	UpdateDivision	Update division details	Success message	✓ Works	Passed
SP-7 9	DeleteDivision	Delete division by ID	Success message	✓ Works	Passed

SP-8 0	GetBatch	Fetch all batches	List of batches	✓ Correct	Passed
SP-8 1	CreateBatch	Insert new batch	Success message + Batch added	✓ Works	Passed
SP-8 2	UpdateBatch	Update batch details	Success message	✓ Works	Passed
SP-8 3	DeleteBatch	Delete batch by ID	Success message	✓ Works	Passed
SP-8 4	GetTimeSlot	Fetch all time slots	List of time slots	✓ Correct	Passed
SP-8 5	CreateTimeSlot	Insert new time slot	Success message + Slot added	✓ Works	Passed
SP-8 6	UpdateTimeSlot	Update time slot	Success message	✓ Works	Passed
SP-8 7	DeleteTimeSlot	Delete time slot by ID	Success message	✓ Works	Passed
SP-8 8	GetMasterTimetable	Fetch complete master timetable	List of all Timetable entries	✓ Correct	Passed

2.3 Sample Data Validation Table

Test Case ID	Table Name	Test Description	Expected Output	Actual Output	Status
SD-01	Users	Verify sample data presence & correctness	Users Data should exist	✓ Data Present, No Issues	✓ Pass
SD-02	Faculty	Verify sample data presence & correctness	Faculty data should exist	✓ Data Present, No Issues	✓ Pass
SD-03	Department	Verify sample data presence & correctness	Department data should exist	✓ Data Present, No Issues	✓ Pass
SD-04	Subject	Verify sample data presence & correctness	Subject data should exist	✓ Data Present, No Issues	✓ Pass
SD-05	Timetable	Verify sample data presence & correctness	Timetable data should exist	✓ Data Present, No Issues	✓ Pass
SD-06	WeekDay	Verify sample data presence & correctness	WeekDay data should exist.	✓ Data Present, No Issues	✓ Pass
SD-07	TimeSlot	Verify sample data presence & correctness	TimeSlot data should exist.	✓ Data Present, No Issues	✓ Pass
SD-08	Role	Verify sample data presence & correctness	Role data should exist	✓ Data Present, No Issues	✓ Pass

SD-09	RolePermission	Verify sample data presence & correctness	RolePermission data should exist	✓ Data Present, No Issues	✓ Pass
SD-10	Permission	Verify sample data presence & correctness	Permissions data should exist	✓ Data Present, No Issues	✓ Pass
SD-11	Batch	Verify sample data presence & correctness	Batch data should exist	✓ Data Present, No Issues	✓ Pass
SD-12	Division	Verify sample data presence & correctness	Divison data should exist	✓ Data Present, No Issues	✓ Pass
SD-13	Location	Verify sample data presence & correctness	Location data should exist	✓ Data Present, No Issues	✓ Pass
SD-14	SubjectType	Verify sample data presence & correctness	Subject Type should exist	✓ Data Present, No Issues	✓ Pass
SD-15	AcademicClass	Verify sample data presence & correctness	AcademicClass data should exist	✓ Data Present, No Issues	✓ Pass
SD-16	AcademicYear	Verify sample data presence & correctness	Academic Year data should exist	✓ Data Present, No Issues	✓ Pass
SD-17	Program	Verify sample data presence & correctness	Program data should exist	✓ Data Present, No Issues	✓ Pass
SD-18	StudentElectiveS ubject	Verify sample data presence & correctness	StudentElectiveSubj ect data should exist	✓ Data Present, No Issues	✓ Pass

SD-19	UserRole	Verify sample data presence & correctness	Userrole data should exist	✓ Data Present, No Issues	✓ Pass
SD-20	StaffDetails	Verify sample data presence & correctness	StaffDetails data should exist	✓ Data Present, No Issues	✓ Pass
SD-21	StudentDetails	Verify sample data presence & correctness	StudentDetails data should exist	✓ Data Present, No Issues	✓ Pass
SD-22	ElectiveSubject	Verify sample data presence & correctness	Elective Subject data should exist	✓ Data Present, No Issues	✓ Pass
SD-23	SubjectCategory	Verify sample data presence & correctness	Subject Category data should exist	✓ Data Present, No Issues	✓ Pass
SD-24	SubjectTeacher	Verify sample data presence & correctness	Subject Teacher data should exist	✓ Data Present, No Issues	✓ Pass

2.4 Database Integrity Testing

Test Case ID	Constraint Type	Table Name	Column(s) Tested	Expected Behavior	Actual Output	Status
DI-01	Primary Key Validation	Users	UserId	Each UserId must be unique & not NULL	✓ No duplicates found	✓ Pass
DI-02	Primary Key Validation	Roles	RoleId	Each RoleId must be unique & not NULL	✓ No duplicates found	✓ Pass
DI-03	Primary Key Validation	Faculties	FacultyId	Each FacultyId must be unique & not NULL	✓ No duplicates found	✓ Pass
DI-04	Primary Key Validation	Departments	DepartmentId	Each DepartmentId must be unique & not NULL	✓ No duplicates found	✓ Pass
DI-05	Foreign Key Validation	Users	RoleId → Role.RoleId	Users.RoleId must exist in Roles table	✓ All records valid	✓ Pass
DI-06	Foreign Key Validation	Users	FacultyId → Faculty.FacultyId	Users.FacultyId must exist in Faculties table	✓ All records valid	✓ Pass
DI-07	Foreign Key Validation	Users	DepartmentId → Department.DepartmentId	Users.DepartmentId must exist in Departments table	✓ All records valid	✓ Pass
DI-08	Foreign Key Validation	Timetable	TimeSlotId → TimeSlot.TimeSlotId	Timetable must reference valid TimeSlots	✓ All records valid	✓ Pass

DI-09	Unique Constraint Validation	Users	Email	Each Email must be unique	✓ No duplicates found	✓ Pass
DI-10	Unique Constraint Validation	Users	Phone Number	Each PhoneNumber must be unique	✓ No duplicates found	✓ Pass
DI-11	NOT NULL Validation	Users	FullName	FullName should not be NULL	✓ No NULL values	✓ Pass
DI-12	NOT NULL Validation	Subjects	Subject Name	SubjectName should not be NULL	✓ No NULL values	✓ Pass

3 API Testing (Backend Testing using Postman)

This section focuses on backend testing of the Timetable Management System using Postman. It verifies whether the APIs perform as expected for operations like login, data insertion, retrieval, and error handling. Each test checks response codes, messages, status (200/401/403/500), and backend behavior for both valid and invalid inputs. Screenshots are included to support the results

All tested APIs are integrated with database operations and stored procedures.

3.1 Authentication & Authorization

Introduction : This section validates the authentication and authorization mechanisms of the TTM API. It ensures that users can log in with valid credentials to obtain a token and that role-based access control (RBAC) restricts endpoints to authorized roles only

Test Cases for 3.1

TC ID	Scenario	API Endpoint	Body/Params	Headers	Expected Response	Actual Response	Status
AUTH-01	Valid login	POST /api/Auth/login	{"email": "vivek.parmar@msubaroda.ac.in", "password": "admin123"}	None	Token (200 OK)	Token returned	Pass
AUTH-02	Invalid credentials	POST /api/Auth/login	{"email": "wrong@msubaroda.ac.in", "password": "wrong"}	None	401 Unauthorized	Error returned	Pass
AUTH-03	Access with valid token	GET /api/Academic/departments	facultyId=1	Authorization: Bearer <token>	List of departments (200 OK)	Data returned	Pass
AUTH-04	Access without token	GET /api/Academic/departments	facultyId=1	None	401 Unauthorized	Error returned	Pass
AUTH-05	Access with invalid token	GET /api/Academic/departments	facultyId=1	Authorization: Bearer wrong-token	401 Unauthorized	Error returned	Pass

3.2 CRUD Operations Testing (Create, Read, Update, Delete)

3.2.1 API Testing - Academic Endpoint :

1. Academic: GET /api/Academic/departments

- **Authorization:** Roles: Admin, Dean, TTM
- **Base URL:** <https://localhost:7073>

TC ID	Scenario	Params	Expected Response	Actual Response	Status
API-01	Valid facultyId	facultyId=1	List of departments (200 OK)	Data returned	Pass
API-02	Invalid facultyId	facultyId=99	Empty [] (200 OK)	No data found	Pass
API-03	Missing facultyId	None	400 Bad Request	Error returned	Pass
API-04	Negative facultyId	facultyId=-1	400 Bad Request	Error returned	Pass
API-05	Non-numeric facultyId	facultyId=abc	400 Bad Request	Error returned	Pass

Screenshots:

API-01 (200 OK with data)

Postman interface showing a GET request to `https://localhost:7073/api/Academic/departments?facultyId=1`. The request is successful, returning a 200 OK status. The response body is JSON, showing a list of departments for facultyId 1.

Query Params

Key	Value	Description
facultyId	1	

Body

```
{
  "departmentId": 1,
  "departmentName": "CSE"
}
```

200 OK • 2.05 s • 603 B

API-02 (200 OK with empty array)

Postman interface showing a GET request to `https://localhost:7073/api/Academic/departments?facultyId=99`. The request is successful, returning a 200 OK status. The response body is JSON, showing an empty array.

Query Params

Key	Value	Description
facultyId	99	

Body

```
[]
```

200 OK • 74 ms • 150 B

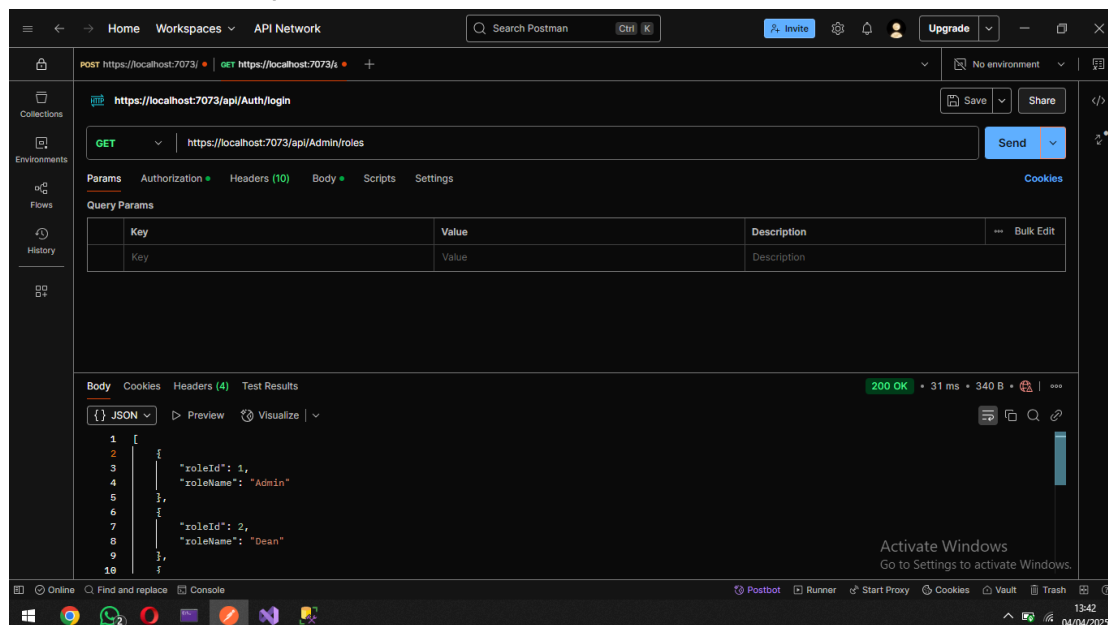
2. Admin: GET /api/Admin/roles

- **Authorization:** Role: Admin
- **Base URL:** <https://localhost:7073>

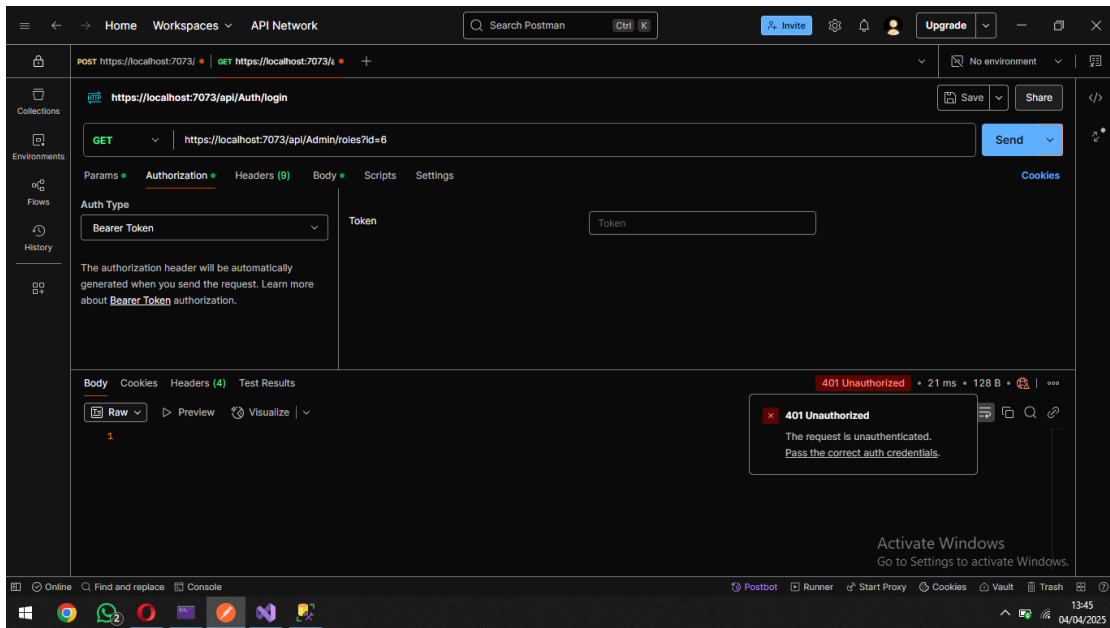
TC ID	Scenario	Headers	Expected Response	Actual Response	Status
API-06	Valid request with token	Authorization: Bearer <token>	List of roles (200 OK)	Data returned	Pass
API-07	No token (Anonymous)	None	401 Unauthorized	Error returned	Pass
API-08	Invalid token	Authorization: Bearer wrong-token	401 Unauthorized	Error returned	Pass
API-09	Extra query params	?id=1 (with valid token)	List of roles (200 OK)	Data returned	Pass
API-010	Wrong endpoint	/api/Admin/role (with token)	404 Not Found	Error returned	Pass

Screenshots:

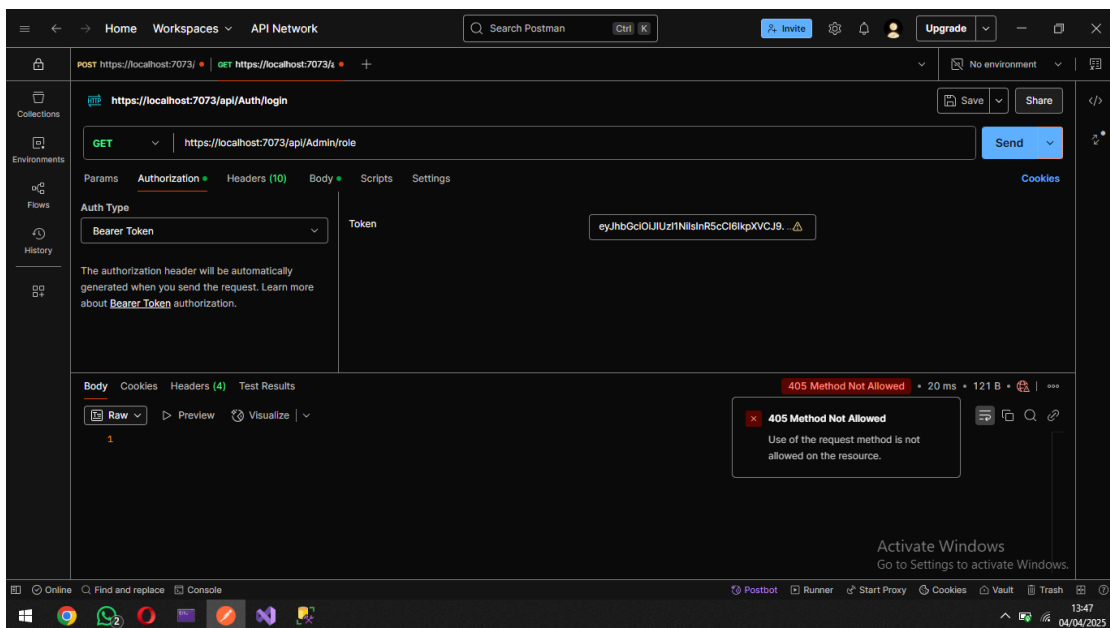
API-06: 200 OK response with roles list.



API-07: 401 Unauthorized error.



API-10: 404 Not Found.



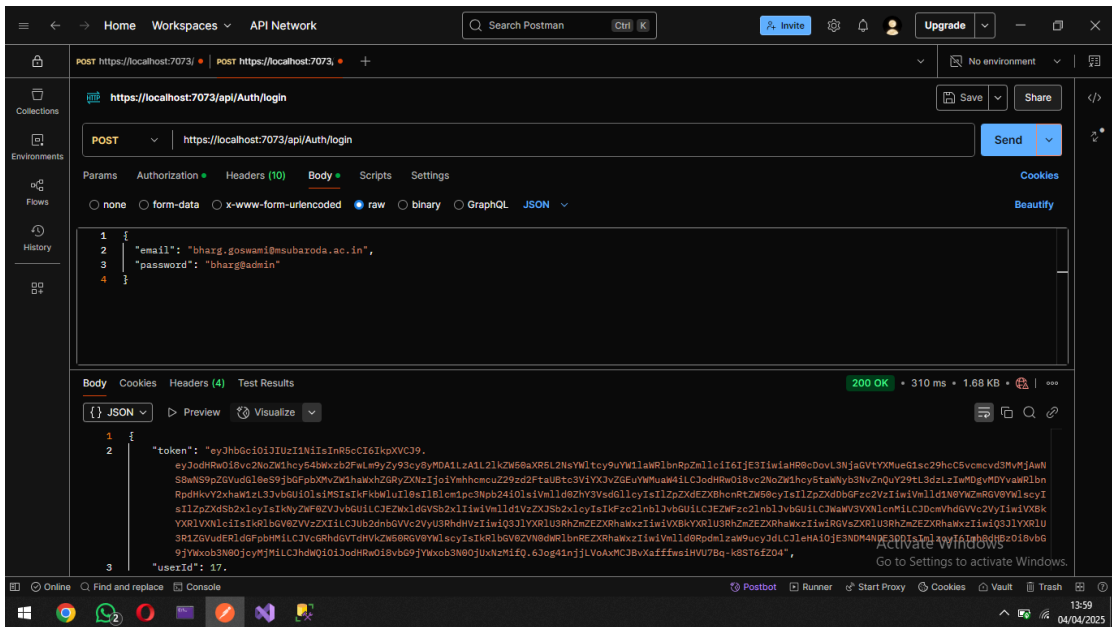
3. Auth: POST /api/Auth/login

- **Authorization:** AllowAnonymous
- **Base URL:** https://localhost:7073

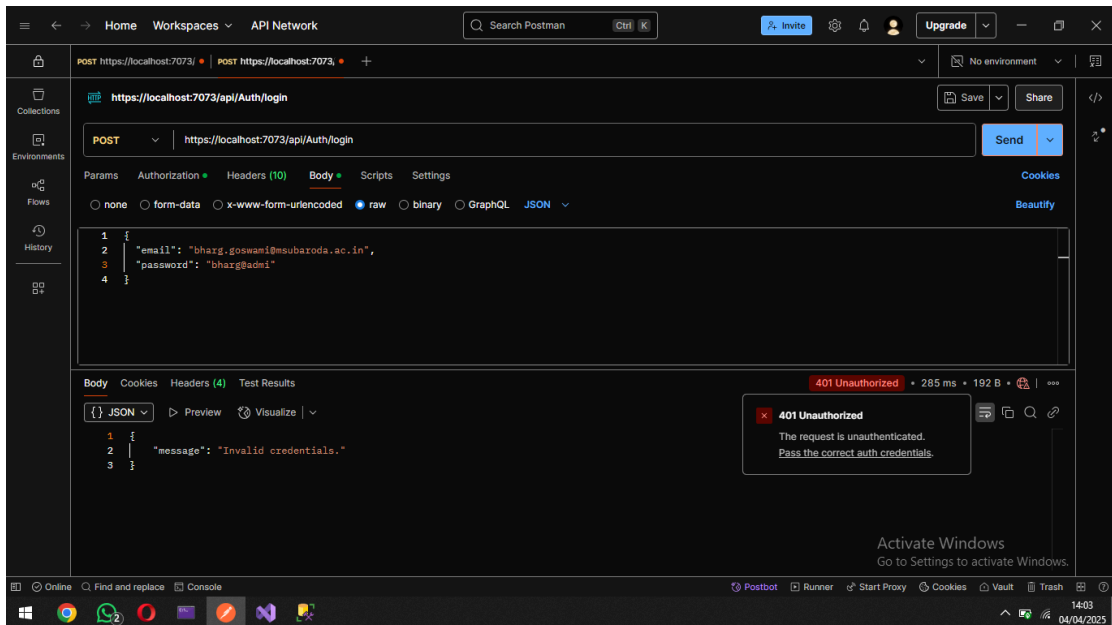
TC ID	Scenario	Body	Expected Response	Actual Response	Status
API-11	Valid credentials	{ "email": "bharg.goswami@msubaroda.ac.in", "password": "bharg@admin" }	Token (200 OK)	Token returned	Pass
API-12	Invalid email	{ "email": "wrong@msubaroda.ac.in", "password": "bharg@admin" }	401 Unauthorized	Error returned	Pass
API-13	Wrong password	{ "email": "bharg.goswmai@msubaroda.ac.in", "password": "wrong" }	401 Unauthorized	Error returned	Pass
API-14	Missing email	{ "password": "admin123" }	400 Bad Request	Error returned	Pass
API-15	Empty body	{ }	400 Bad Request	Error returned	Pass

Screenshots:

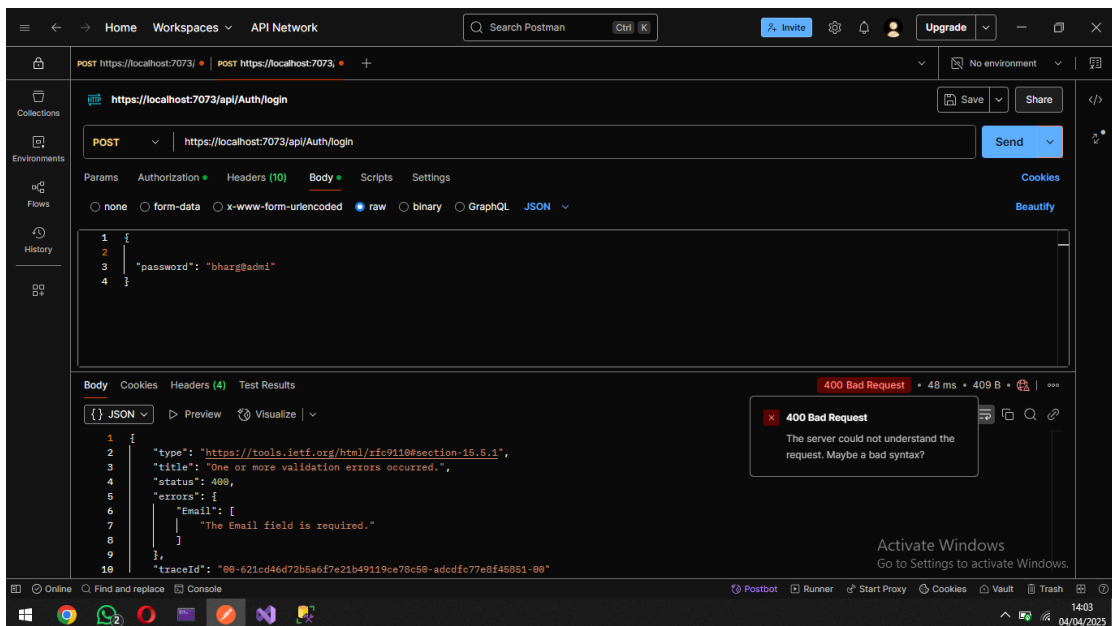
API-11 (200 OK with token)



API-13 (401 Unauthorized)



API-14 (400 Missing Email)

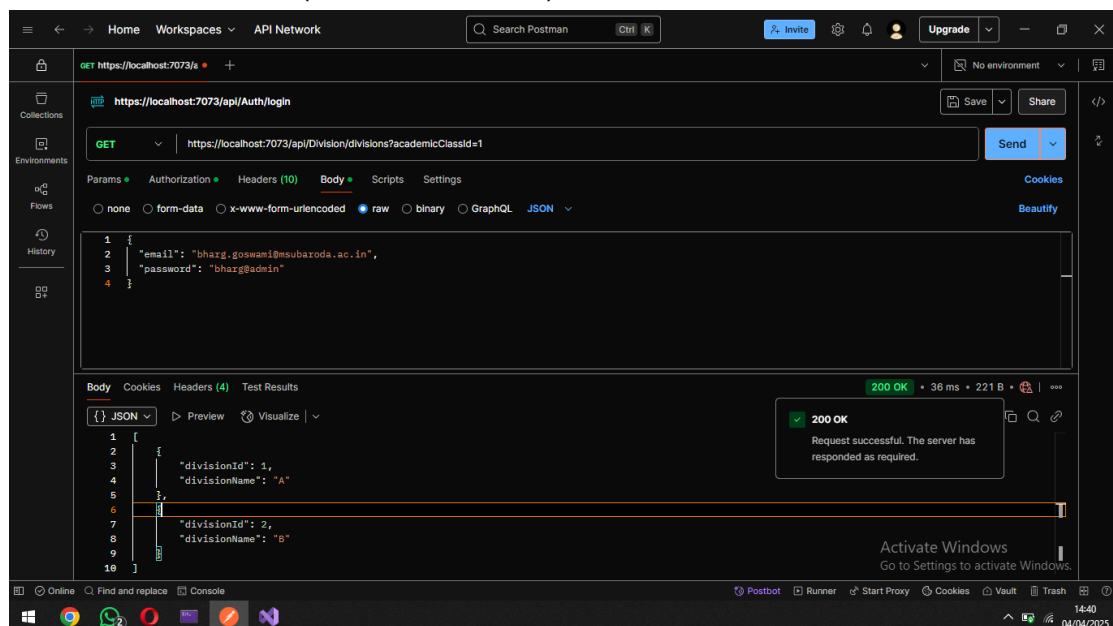


4. Division: GET /api/Division/divisions

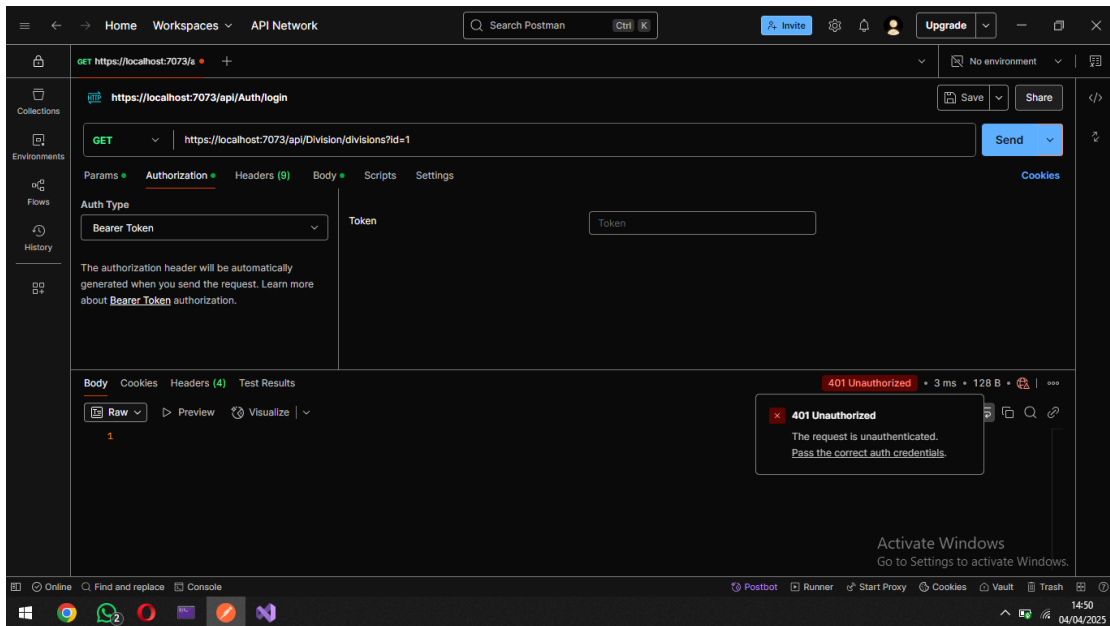
- **Authorization:** To All Role
- **Base URL:** https://localhost:7073

TC ID	Scenario	Params	Headers	Expected Response	Actual Response	Status
API-16	Valid request with token	academicClassId=1	Authorization : Bearer <token>	List of divisions (200 OK)	Data returned	Pass
API-17	No token	academicClassId=1	None	401 Unauthorized	Error returned	Pass
API-18	Invalid endpoint	Wrong URL	Authorization : Bearer <token>	404 Not Found	Error returned	Pass
API-19	Missing academicClassId	None	Authorization : Bearer <token>	400 Bad Request or Empty []	Error/Empty	Pass
API-20	Invalid academicClassId	academicClassId=99	Authorization : Bearer <token>	Empty [] (200 OK) or 404	No data found	Pass

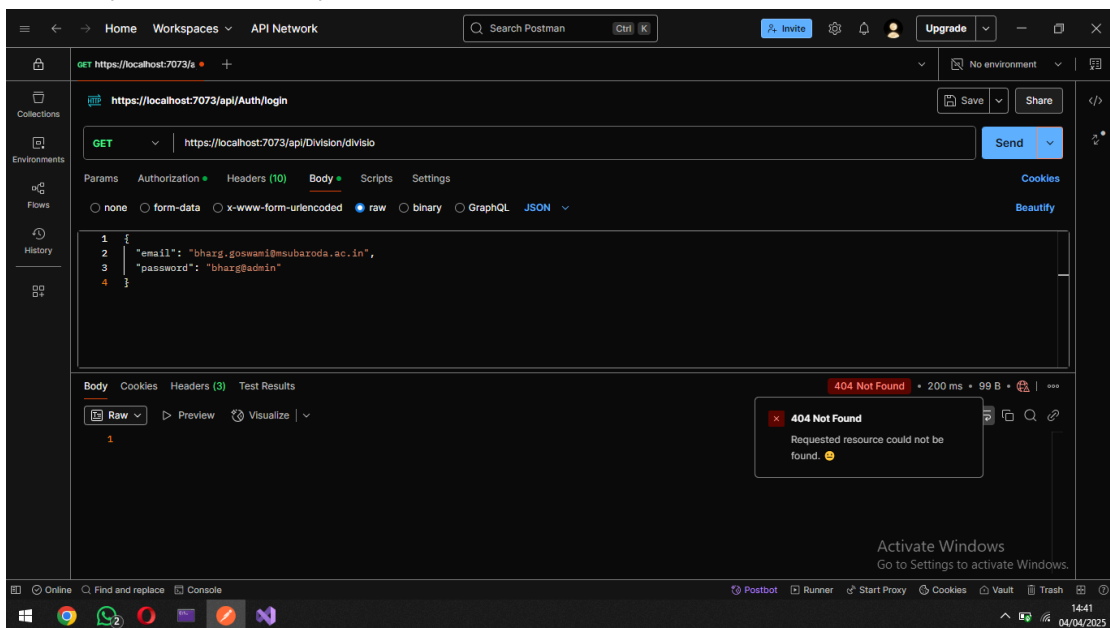
Screenshots: API-16 (200 OK with data)



API-17 (W.O TOKEN 401 Not Found)



API-18 (404 Not Found)



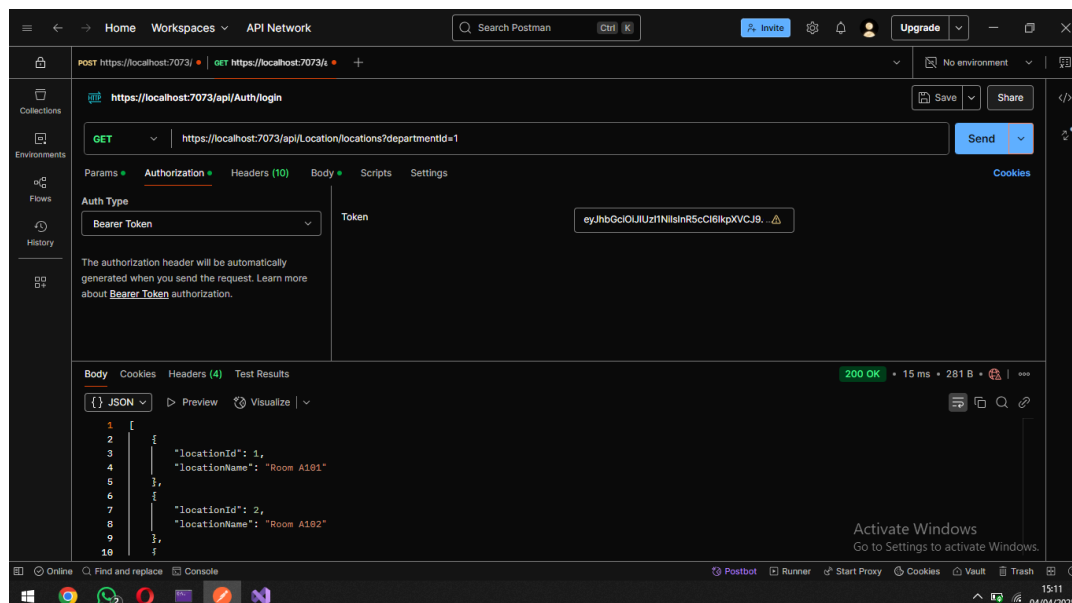
5. Location: GET /api/Location/locations

- **Authorization:** Roles: TTM, Dean, HOD, Teacher
- **Base URL:** https://localhost:7073

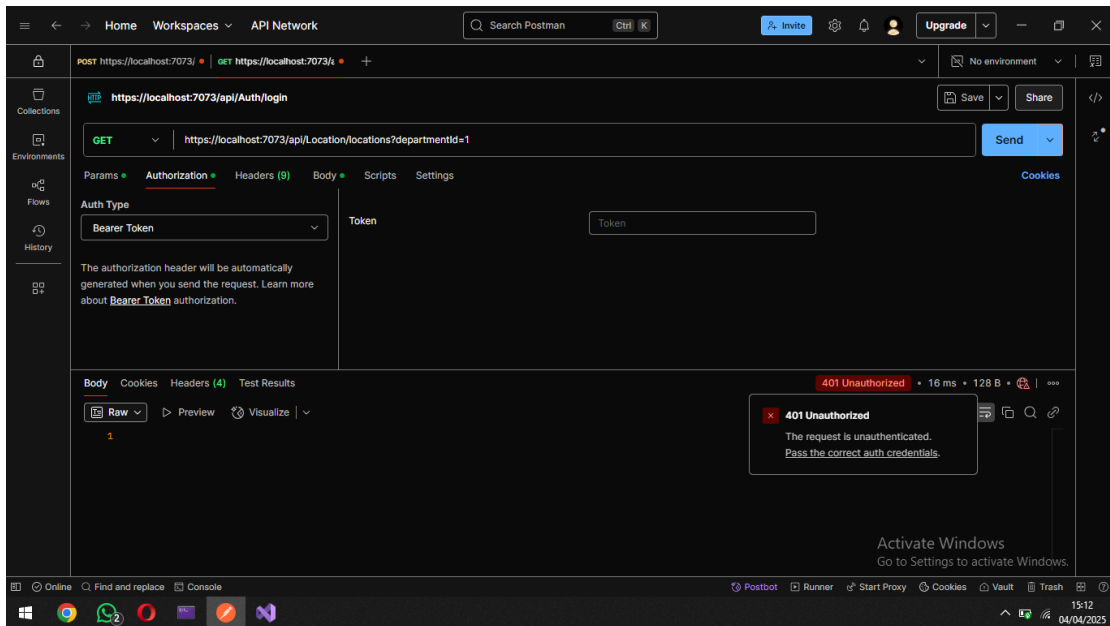
TC ID	Scenario	Params	Headers	Expected Response	Actual Response	Status
API-21	Valid request with token(Ex: TTM)	departmentId=1	Authorization: Bearer <token>	List of locations (200 OK)	Data returned	Pass
API-22	No token	departmentId=1	None	401 Unauthorized	Error returned	Pass
API-23	Invalid endpoint	Wrong URL	Authorization: Bearer <token>	404 Not Found	Error returned	Pass
API-24	Missing departmentId	None	Authorization: Bearer <token>	400 Bad Request or Empty Response	Error/Empty	Pass
API-25	Server down (simulate)	None	None	500 Internal Server Error	Error returned	Pass

Screenshots:

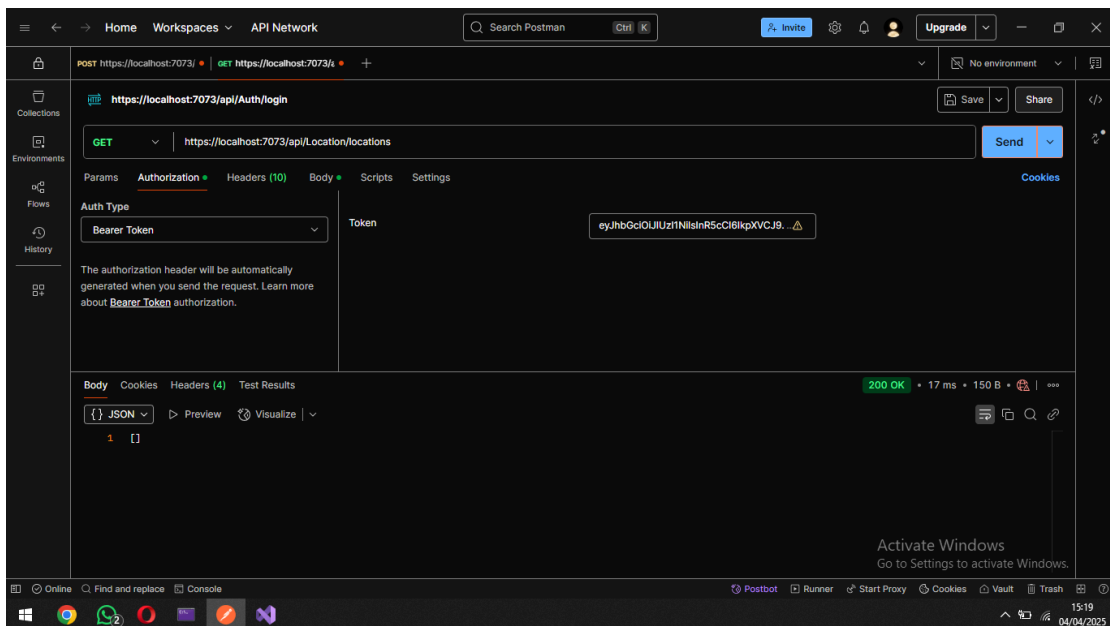
API-21 (200 OK with data)



API-22 (401 Unauthorized):



API-24 (400 Unauthorized):



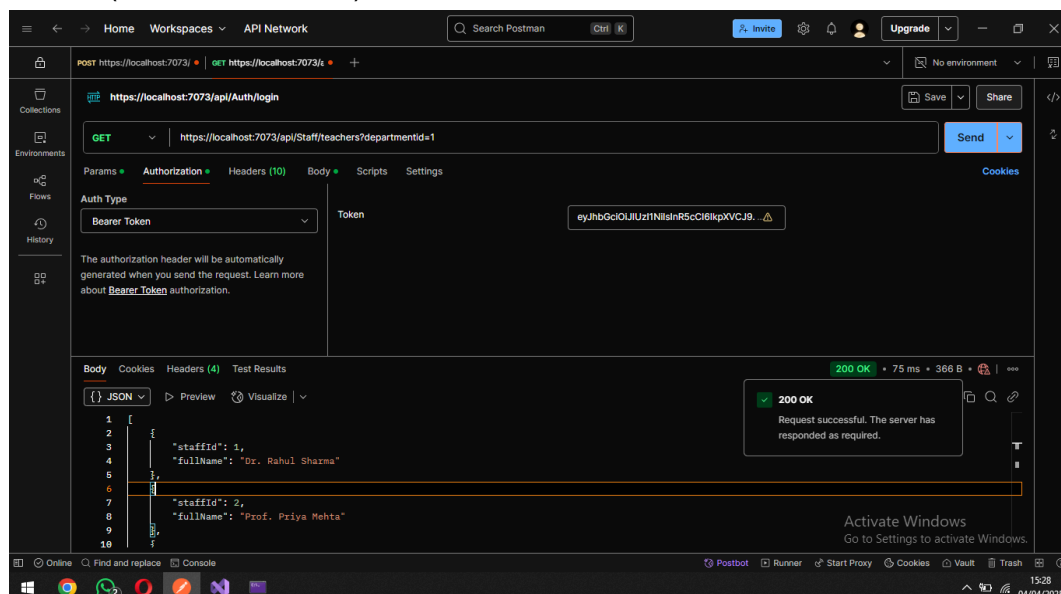
6. Staff: GET /api/Staff/teachers

- **Authorization:** Roles: TTM, Dean, HOD, Teacher
- **Base URL:** <https://localhost:7073>

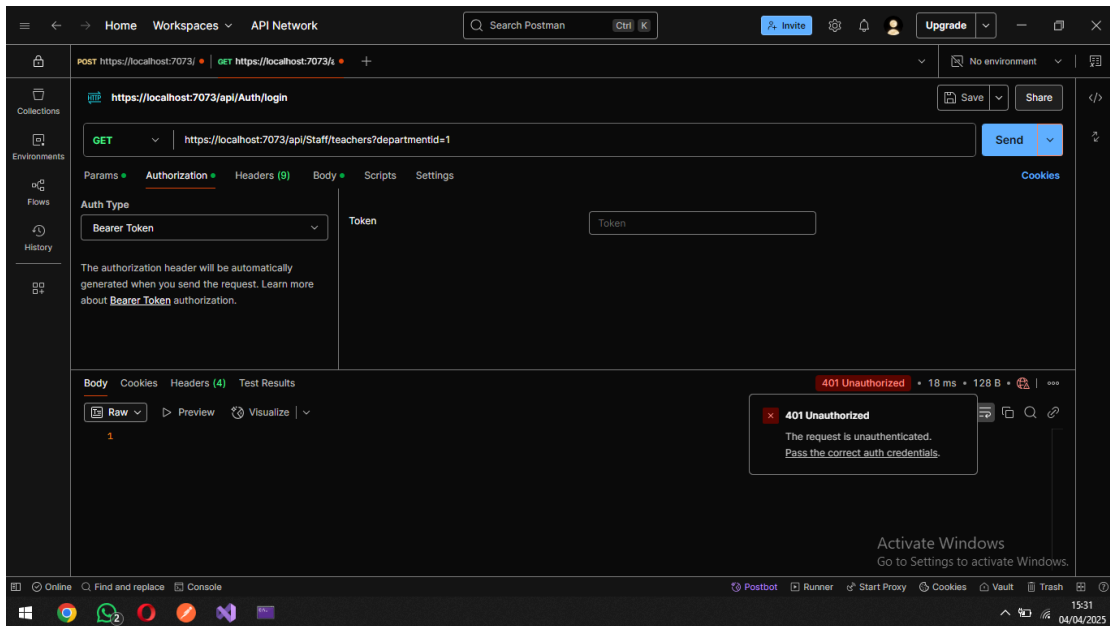
TC ID	Scenario	Params	Headers	Expected Response	Actual Response	Status
API-26	Valid request with token	departmentId=1	Authorization: Bearer <token>	List of teachers (200 OK)	Data returned	Pass
API-27	No token	departmentId=1	None	401 Unauthorized	Error returned	Pass
API-28	Invalid endpoint	Wrong URL	Authorization: Bearer <token>	404 Not Found	Error returned	Pass
API-29	Missing departmentId	None	Authorization: Bearer <token>	400 Bad Request or Empty Response	Error/Empty	Pass
API-30	Invalid departmentId	departmentId=99	Authorization: Bearer <token>	Empty [] (200 OK) or 404	No data found	Pass

Screenshots:

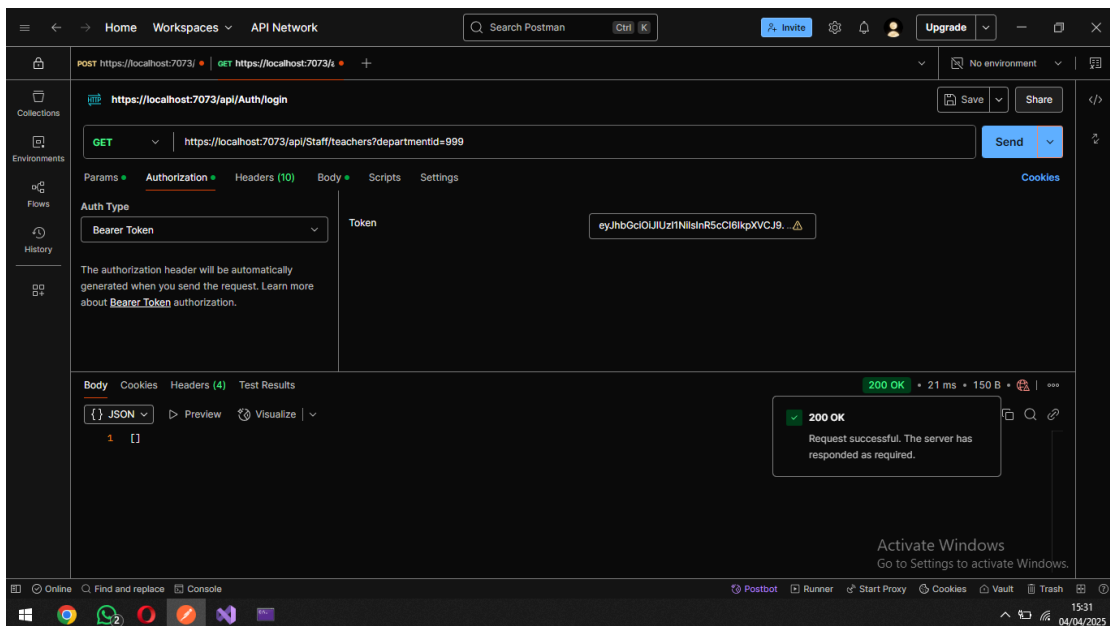
API-26 (200 OK with data)



API-27 (401 Unauthorized)



API-30 (200 Empty)



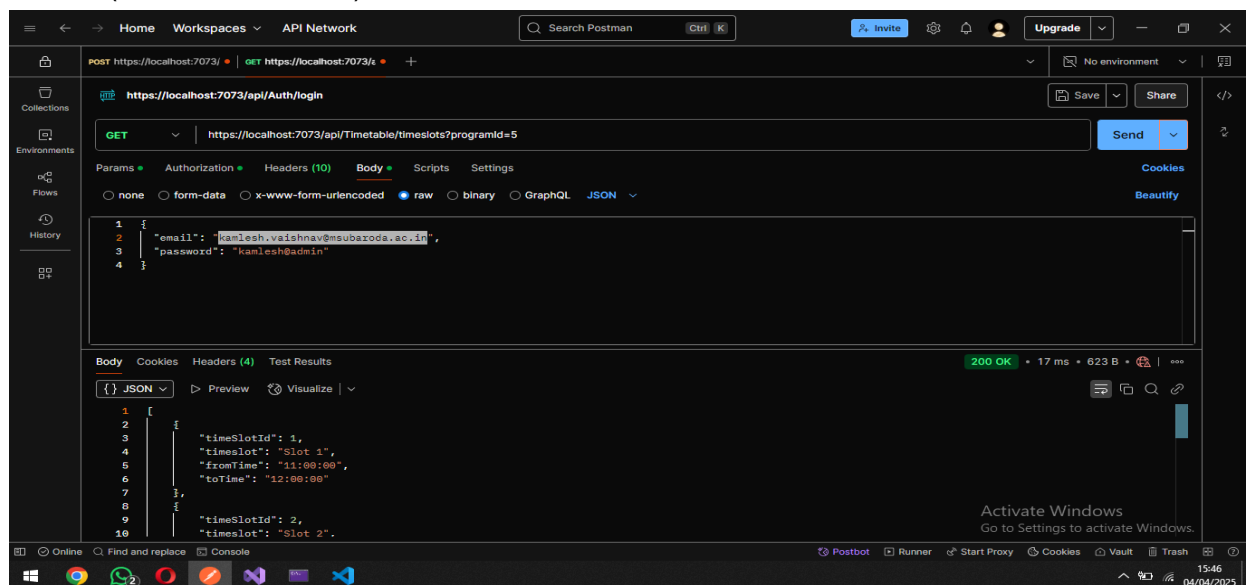
7. Timetable: GET /api/Timetable/timeslots

- **Authorization:** Roles: TTM, Dean, HOD, Teacher, Student
- **Base URL:** <https://localhost:7073>

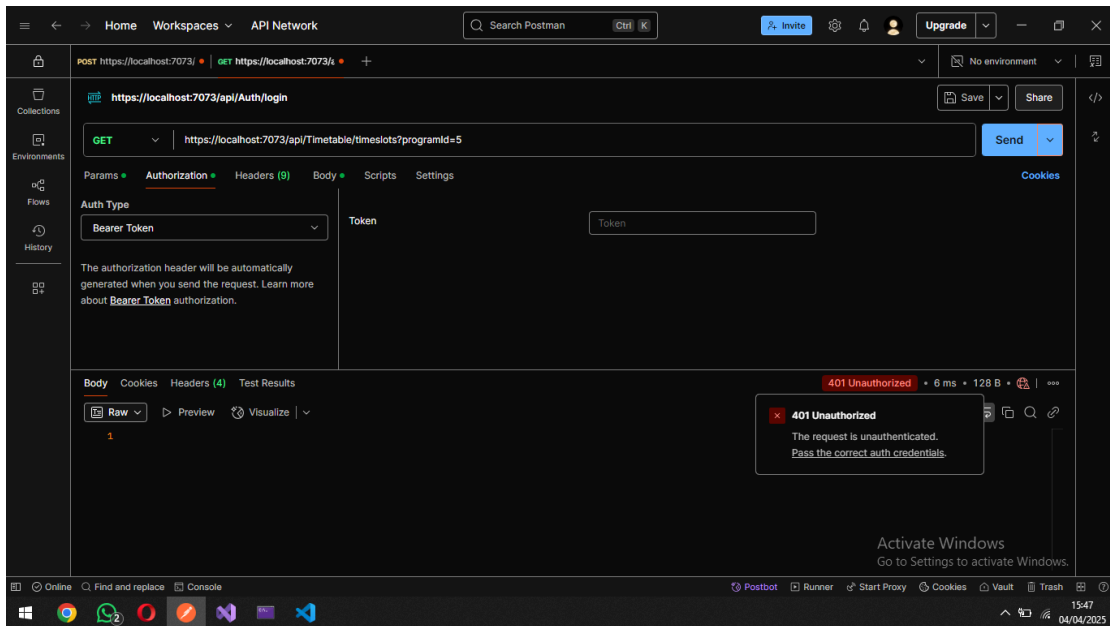
TC ID	Scenario	Params	Headers	Expected Response	Actual Response	Status
API-31	Valid request with token	program id=1	Authorization: Bearer <token>	Timeslots list/details (200 OK)	Data returned	Pass
API-32	No token	timeslotId=1	None	401 Unauthorized	Error returned	Pass
API-33	Invalid endpoint	Wrong URL	Authorization: Bearer <token>	404 Not Found	Error returned	Pass
API-34	Missing timeslotId	None	Authorization: Bearer <token>	400 Bad Request or Empty Response	Error/Empty	Pass
API-35	Invalid timeslotId	timeslotId=99	Authorization: Bearer <token>	Empty [] (200 OK) or 404	No data found	Pass

Screenshots:

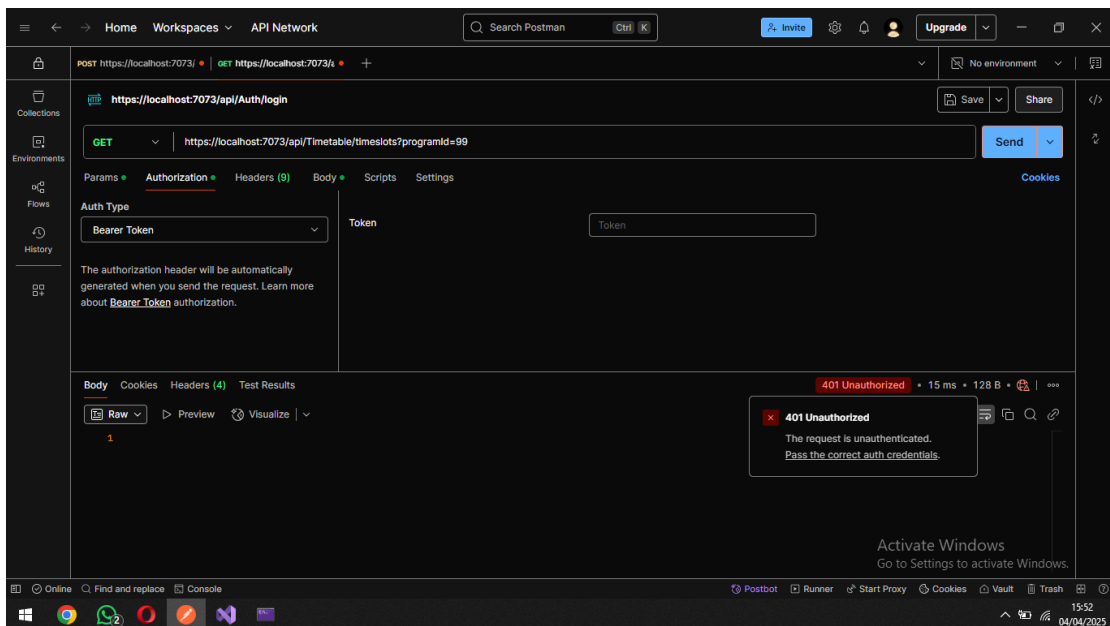
API-31 (200 OK with data)



API-32 (401 Unauthorized)



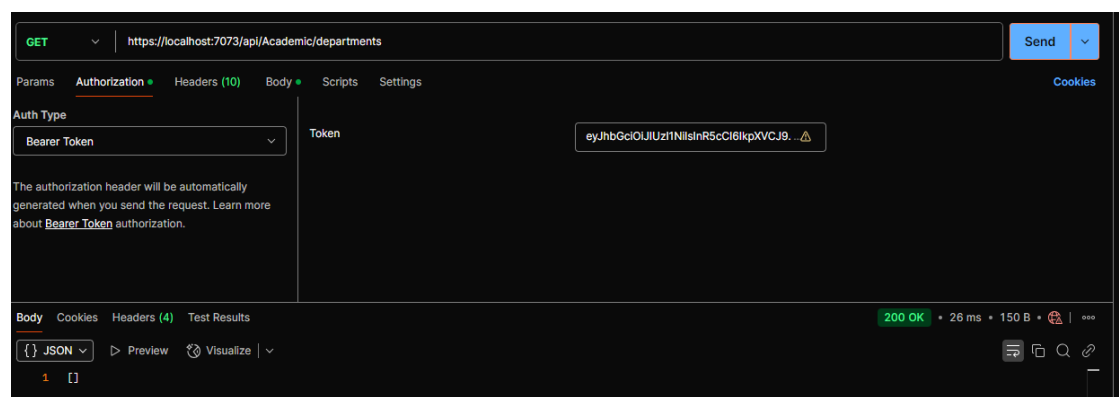
API-35 (401 Unauthorised)



3.3 Error Handling & Validation

TC ID	API Endpoint	Scenario	Input	Expected Response	Status
API-ERR-01	/api/Academic/departments	Missing required query param	No facultyId	Return 400 Bad Request	✗ Failed
API-ERR-02	/api/Admin/role/assign	Invalid userId/roleId	userId = 9999, roleId = 9999	400 Bad Request – "Error: User not found or is deleted."	✓ Pass
API-ERR-03	/api/Auth/login	Wrong login credentials	Wrong email or password	401 Unauthorized + "User not found." message	✓ Pass
API-ERR-04	/api/Admin/user (POST)	Missing required fields (email, password)	Only FullName provided	400 Bad Request – Validation errors	✓ Pass
API-ERR-05	/api/Academic/subjects	Unauthorized access	No token	401 Unauthorized	✓ Pass

ScreenShots :
API-ERR-01



Note: API accepts missing facultyId and returns an empty array instead of throwing a validation error. This indicates that backend validation might be skipped

API-ERR-02

POSThttps://localhost:7073/api/Admin/role/assignSend

ParamsAuthorizationHeaders (10)BodyScriptsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

```
1 {
2   "userId": 9999,
3   "roleId": 9999
4 }
5
```

BodyCookiesHeaders (4)Test Results400 Bad Request • 175 ms • 205 B

JSONPreviewVisualize

```
1 {
2   "error": "Error: User not found or is deleted."
3 }
```

API-ERR-03

POSThttps://localhost:7073/api/Auth/login

ParamsAuthorizationHeaders (9)BodyScriptsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

```
1 {
2   "email": "wrong@email.com",
3   "password": "wrongpassword"
4 }
5
```

BodyCookiesHeaders (4)Test Results401 Unauthorized • 27 ms

JSONPreviewVisualize

```
1 {
2   "message": "User not found."
3 }
```

API-ERR-04

The screenshot shows a Postman interface for a POST request to `https://localhost:7073/api/Admin/user`. The request body is a JSON object: `{ "fullName": "Test User" }`. The response is a 400 Bad Request with the following JSON body:

```
{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "Email": [
      "The Email field is required."
    ],
    "Password": [
      "The Password field is required."
    ],
    "PhoneNumber": [
      "The PhoneNumber field is required."
    ]
  },
  "traceId": "00-274ebae03079792abdf6bd221333e347-2d28dbceea818c0d-00"
}
```

The status bar indicates a 400 Bad Request, 137 ms, and 509 B. An "Activate Windows" watermark is visible in the bottom right corner.

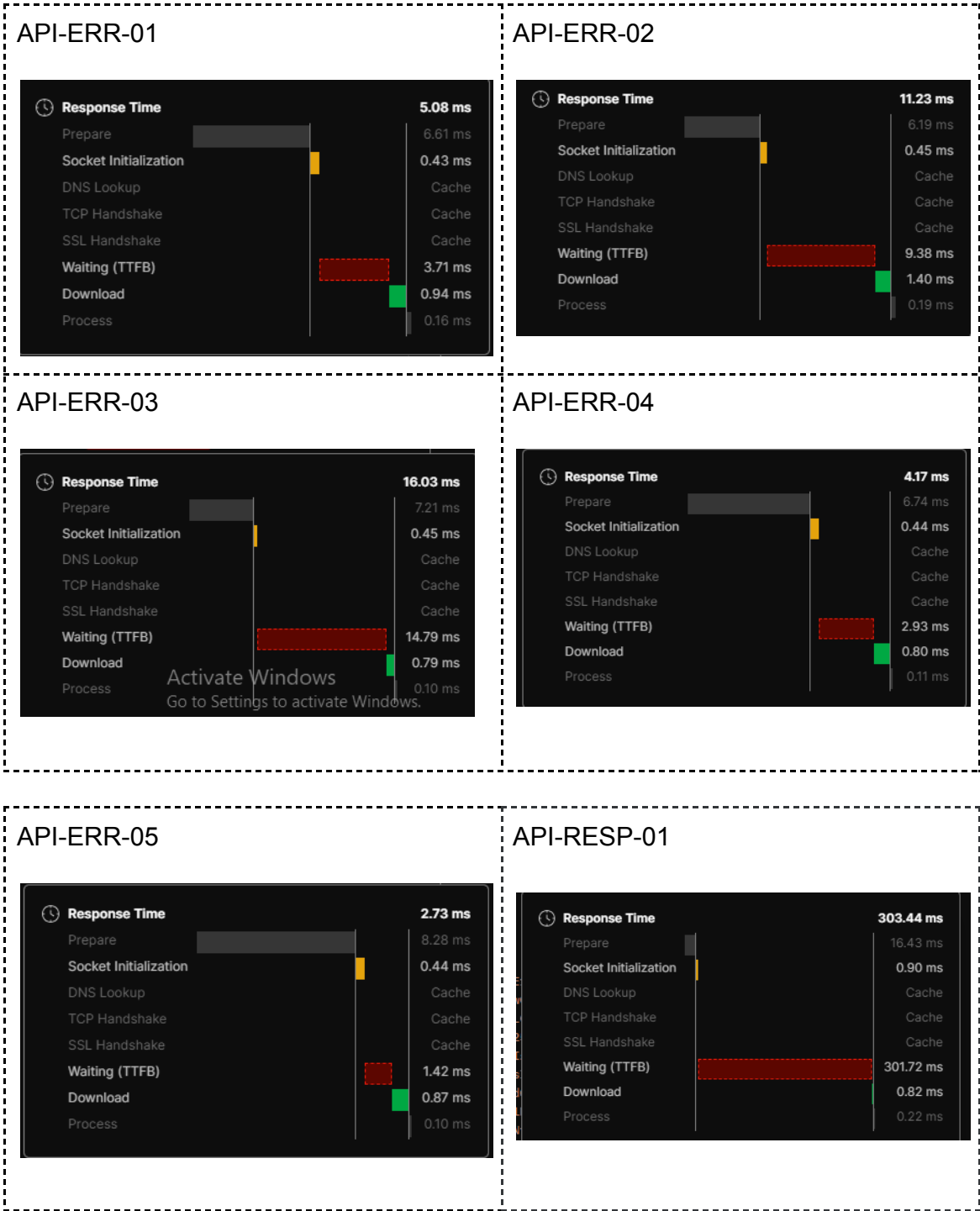
API-ERR-05

The screenshot shows a Postman interface for a GET request to `https://localhost:7073/api/Academic/subjects`. The request is configured with Bearer Token authentication. The response is a 401 Unauthorized with a status bar indicating 21 ms and 128 B. The "Raw" tab is selected in the response pane, showing the raw response text.

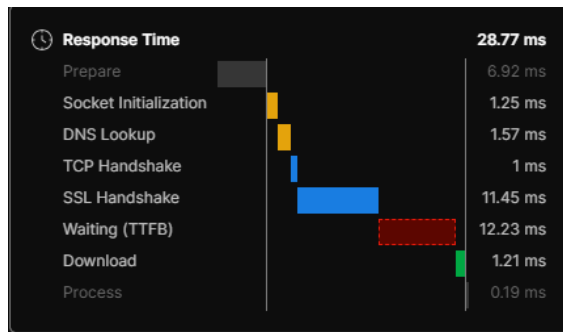
3.4 Response Time & Status Codes

TC ID	API Endpoint	Scenario	Input / Case	Expected Output	Status Code	Response Time
API-ERR-01	/api/Academic/departments	Missing required param	facultyId not passed	[]	200	5.08 ms
API-ERR-02	/api/Admin/role/assign	Invalid userId / roleId	userId = 9999, roleId = 9999	Error: User not found or is deleted.	400	11.23 ms
API-ERR-03	/api/Auth/login	Wrong login credentials	wrong email / password	message: User not found.	401	16.03 ms
API-ERR-04	/api/Admin/user (POST)	Missing required fields	FullName only, no email/password	Validation error or custom message	400	4.17 ms
API-ERR-05	/api/Academic/subjects	Unauthorized access	No token in headers	message: Unauthorized	401	2.73 ms
API-RESP-01	/api/Auth/login	Valid credentials	Correct email & password	Login success or token returned	200	303.44 ms
API-RESP-02	/api/Admin/users	Get list of users	GET with valid token, facultyId=1, departmentId=2	JSON list of users	200	28.77 ms
API-RESP-03	/api/Academic/faculties	Simple GET with no param	GET call	JSON list of faculties	200	42.16 ms

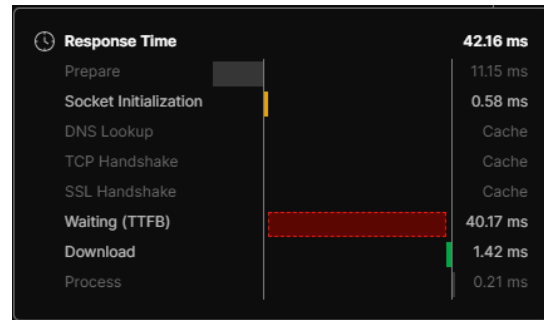
Screenshots :



API-RESP-02



API-RESP-03



4 UI Testing (Frontend Testing)

Introduction: This section focuses on validating the frontend user interface of the Timetable Management System. The goal is to ensure the UI is intuitive, responsive, and functional across key components like login, navigation, forms, and timetable management. Testing will cover usability, form validations, and correct interaction with backend APIs.

4.1 Login Page Testing

TC ID	Scenario	Test Description	Input / Case	Expected Output	Status
UI-LOGI N-01	Valid login	Test login with correct credentials.	Email: bharg.goswami@msub aroda.ac.in, Password: bharg@admin	Redirects to dashboard/homep age, no errors.	Pass
UI-LOGI N-02	Invalid email	Test login with wrong email.	Email: wrong@msubaroda.ac .in, Password: bharg@admin	Error message: Invalid Credentials. Please try again.	Pass
UI-LOGI N-03	Wrong password	Test login with correct email but wrong password.	Email: bharg.goswami@msub aroda.ac.in, Password: wrong	Error message: "Invalid credentials" or similar, stays on login page.	Pass
UI-LOGI N-04	Missing email	Submit form with empty email field.	Email: (blank), Password: bharg@admin	Client-side validation: "Please Fill out this Field", form doesn't submit.	Pass
UI-LOGI N-05	Missing password	Submit form with empty password field.	Email: bharg.goswami@msub aroda.ac.in, Password: (blank)	Client-side validation: "Please Fill out this Field", form doesn't submit.	Pass
UI-LOGI N-06	Empty form submissi on	Submit form with both fields empty.	Email: (blank), Password: (blank)	Client-side validation: "Please Fill out this Field",	Pass

				form doesn't submit.	
UI-LOGI N-07	Invalid email format	Test login with incorrectly formatted email.	Email: invalid.email, Password: bharg@admin	Client-side validation: "Invalid email format", form doesn't submit.	Pass
UI-LOGI N-08	UI responsiveness	Check login page on different screen sizes.	Resize browser: Mobile (375px), Tablet (768px), Desktop (~1200px).	UI elements (form, button, labels) remain usable, no overlap or cutoff.	Pass
UI-LOGI N-09	Network failure	Simulate API failure (e.g., backend down).	Valid credentials, but backend stopped.	Error message: "No response from server. Please check your network connection, stay on login page.	Pass
UI-LOGI N-10	Console errors	Check for JavaScript errors during login attempts.	Valid and invalid login attempts.	No console errors or warnings in browser Developer Tools.	Pass

Screenshots :

(UI-LOGIN-01)

localhost:5173/login

THE MAHATMA SAVITRIBAI PHULE UNIVERSITY OF BARODA
सत्यं शिवं सुन्दरम्
Estd. 1949

Login

Email

bharg.goswami@msubaroda.ac.in

Password

.....

Login

Activate Windows
Go to Settings to activate Windows

THE MAHATMA SAVITRIBAI PHULE UNIVERSITY OF BARODA

Role: Admin Logout

Role Management

CREATE NEW ROLE

Sr No.	Role Name	Permissions	Actions
1	Admin	- ViewFaculties - ViewDepartments - ViewClasses - ViewStaffDetails - ViewRolesPermissions - CreateRole - DeleteRole - ViewUserRoles - AssignRole - DeassignRole	 

(UI-LOGIN-02 & 03)

Invalid Credentials. Please try again.



सत्यं शिवं सुन्दरम्
Estd. 1949

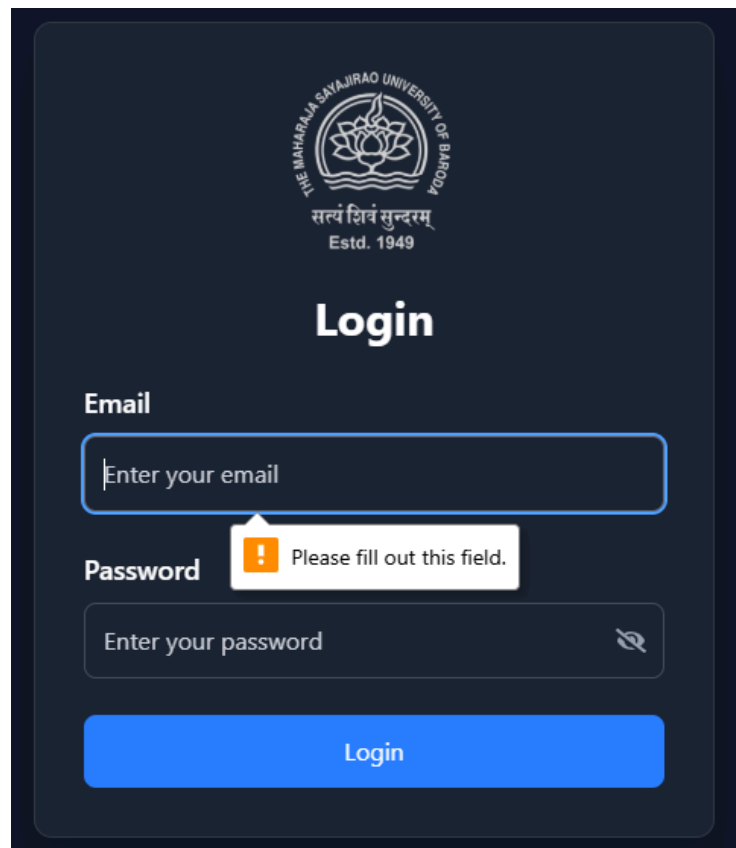
Login

Email

Password

Login

UI-LOGIN-06



The image shows a login form for The Maharaja Sayajirao University of Baroda. At the top is the university's logo, which includes a circular emblem with a lotus and the text 'THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA' and 'सत्यं शिवं सुन्दरम्' (Satyam Shivam Sundaram) in Sanskrit, with 'Estd. 1949' below it. The word 'Login' is centered in a large, bold font. Below it are two input fields: 'Email' and 'Password'. The 'Email' field contains the placeholder text 'Enter your email'. The 'Password' field contains the placeholder text 'Enter your password' and has a toggle icon (an eye with a slash) on its right side. A blue 'Login' button is positioned below the password field. A white error message box with an orange exclamation mark icon is displayed above the password field, stating 'Please fill out this field.'

THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA
सत्यं शिवं सुन्दरम्
Estd. 1949

Login

Email

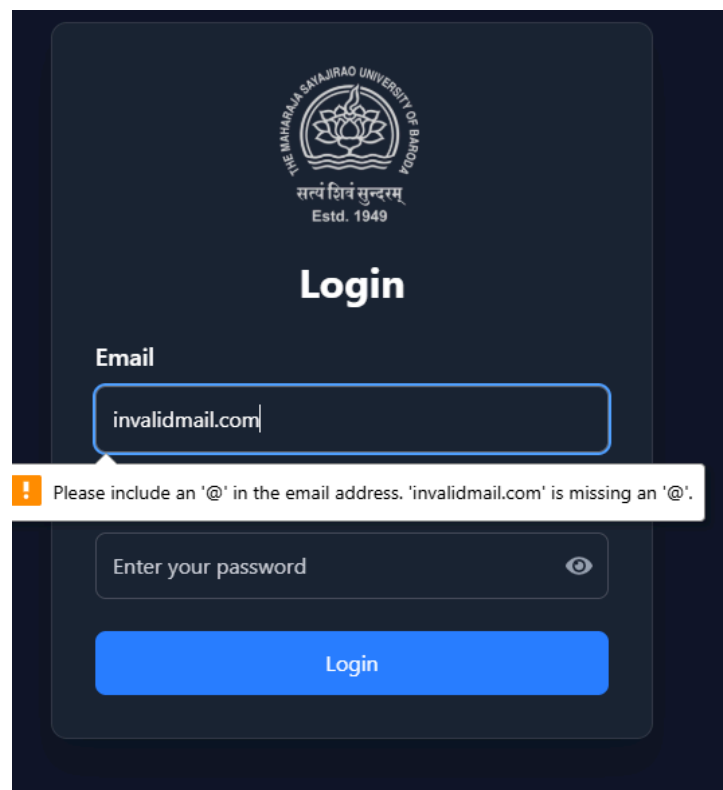
Enter your email

Password

Enter your password

Login

! Please fill out this field.



This image shows the same login form as the previous one, but with the 'Email' field filled with the text 'invalidmail.com'. A white error message box with an orange exclamation mark icon is now positioned below the email field, stating 'Please include an '@' in the email address. 'invalidmail.com' is missing an '@'.

THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA
सत्यं शिवं सुन्दरम्
Estd. 1949

Login

Email

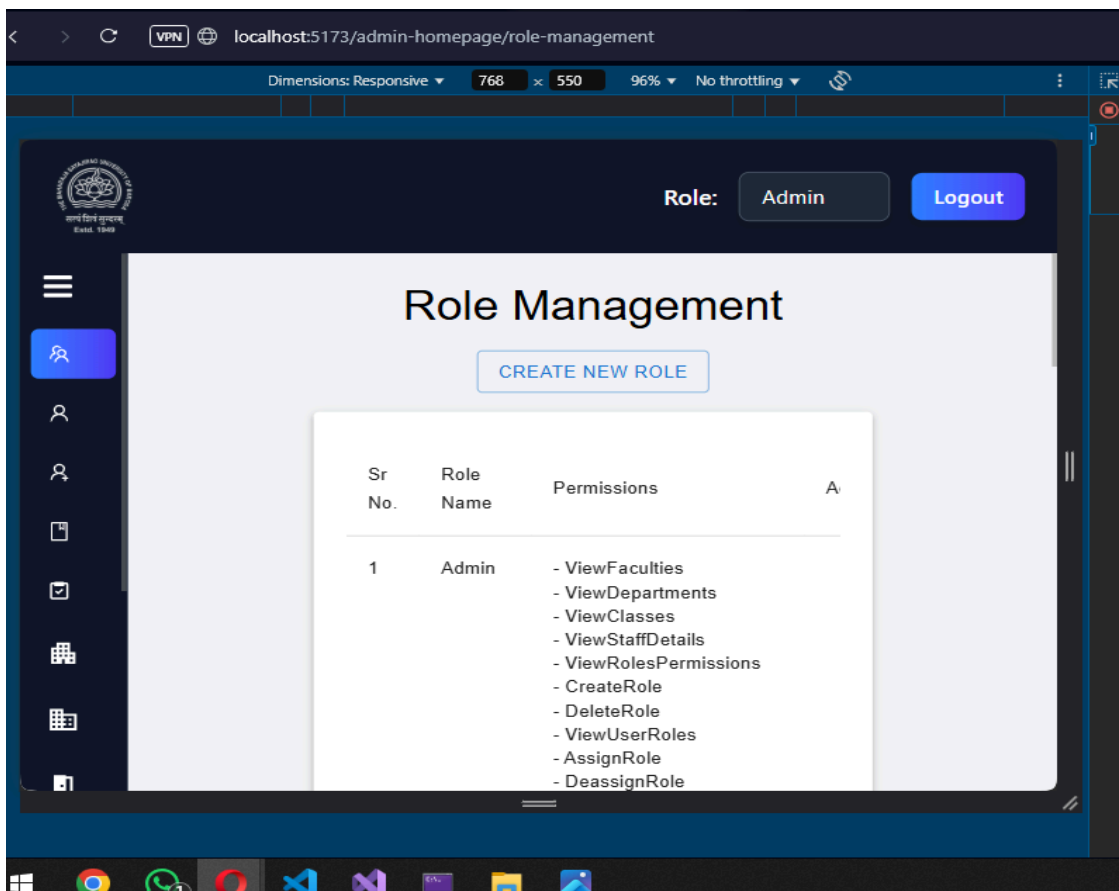
invalidmail.com

Enter your password

Login

! Please include an '@' in the email address. 'invalidmail.com' is missing an '@'.

UI-LOGIN-08 Responsive Design



4.2 Navigation Testing

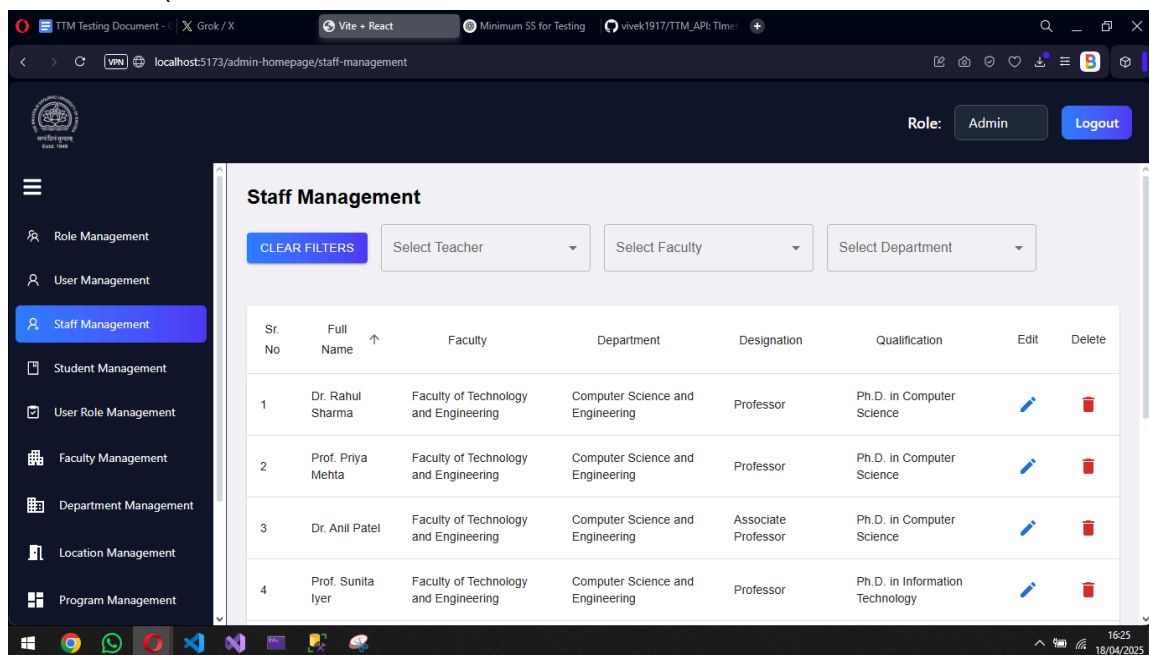
TC ID	Scenario	Test Description	Input / Case	Expected Output	Status
UI-NAV-01	Admin accesses all menu links	Test all menu links for Admin role.	Log in as Admin, click each link (Dashboard, Users, Timetable, Roles).	All links redirect to correct pages, 200 OK for API calls.	Passed
UI-NAV-02	Student accesses restricted page	Test if Student can access Admin-only page (e.g., Users).	Log in as Student, click Users link or enter URL.	Student is redirected to <code>/login</code> page upon accessing <code>/users</code> .	Pass
UI-NAV-03	Broken link handling	Test a non-existent link or wrong URL.	Enter <code>http://localhost:5173/invalid</code> or click broken link.	User is redirected to login page when accessing invalid URL. No crash.	Pass
UI-NAV-04	Back/Forward navigation	Test browser Back/Forward buttons after navigating.	Log in, navigate Dashboard → Users → Timetable, use Back/Forward.	Pages load correctly, state preserved (e.g., filters intact).	Failed
UI-NAV-05	Role-based menu visibility	Verify menu items shown/hidden based on role.	Log in as Admin vs. Student, check menu.	Admin sees all links (Users, Roles); Student sees limited (Timetable).	Pass
UI-NAV-06	Mobile responsiveness	Test navigation on mobile view.	Resize to ~375px, toggle menu, click links.	Menu collapses (e.g., hamburger), links accessible, no overlap.	Pass

UI-NAV-07	Page load time	Measure load time for key pages.	Navigate to Dashboard, Users, Timetable, check Network tab.	Pages load <2s, API calls <500ms	Pass
UI-NAV-08	Console errors	Check for JS errors during navigation.	Navigate all links as Admin, check Console tab.	No errors/warnings in Console.	Pass
UI-NAV-09	Network failure	Test navigation when backend is down.	Stop backend, navigate to Users or Timetable.	No response from server. Please check your network connection.	Pass
UI-NAV-10	Logout navigation	Test logout functionality.	Log in, click Logout, check redirect.	Redirect to login confirmed & Protected pages not accessible without login	Pass

ScreenShots :

Figure 1: Admin Menu and Dashboard

Load time <2s (UI-NAV-01, UI-NAV-05, UI-NAV-07, UI-NAV-08)



Student menu with only Timetable link. Restricted pages not visible.Covers: UI-NAV-02, UI-NAV-05.

[illegible]

Figure 3: Mobile View of Menu(UI-NAV-06)

Dimensions: iPhone 14 Pro Max ▾ 430 x 932 50% ▾ No throttling ▾ (10)

University of Technology and Engineering

Role: Student Logout

☰

⏻

Timetable

Full Name: Amit Prajapati **Faculty:** Faculty of Technology and Engineering

Class: MCA - Semester 2 **Department:** Computer Science and Engineering

Division: C **Program:** Master of Computer Applications

Friday	Saturday
-	-
-	-
-	-
-	-
-	-
-	-
-	-
-	-

Artificial Intelligence
Room B201
Dr. Anil Patel

1. SUBJECT 2. BATCH 3. LOCATION 4. TEACHER

4.3 Form Validations

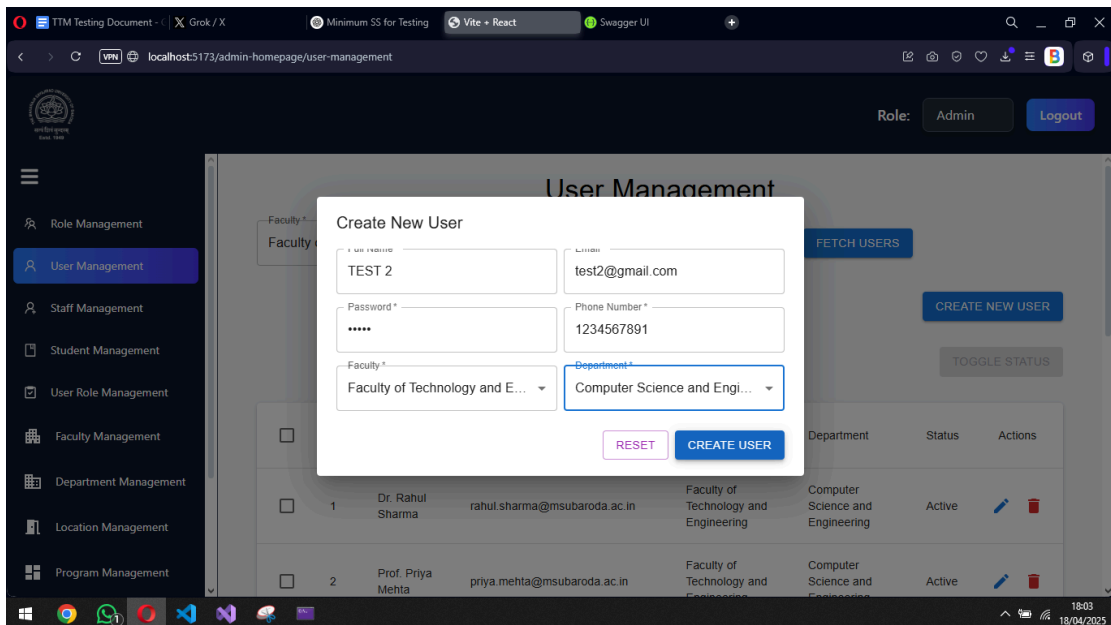
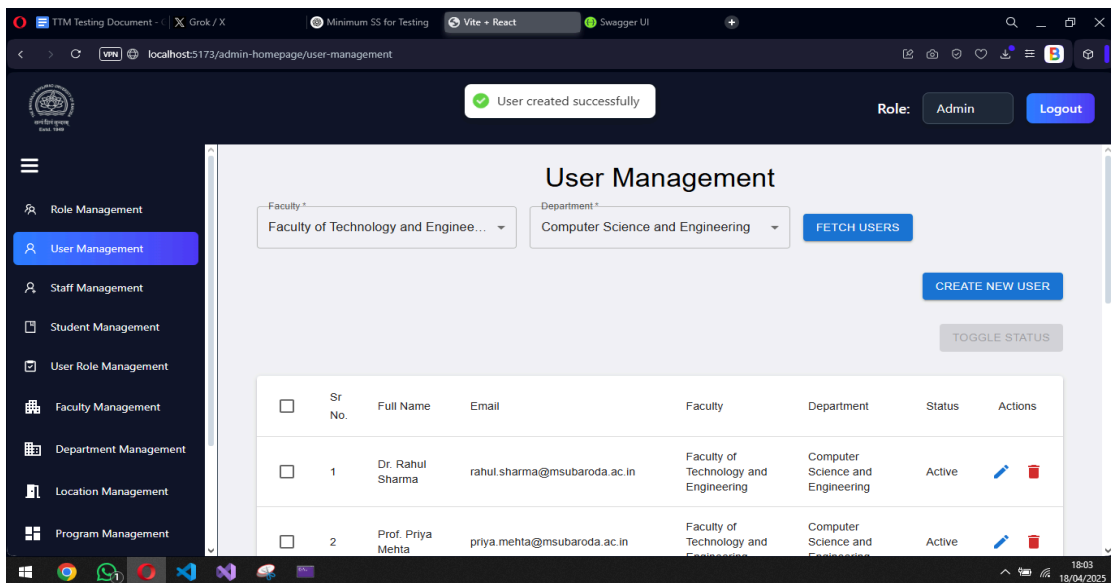
Test Case Table

TC ID	Scenario	Test Description	Input/Case	Expected Output	Status
UI-FO RM-01	Admin creates user with valid inputs	Login as Admin, submit Create User form with valid data	Email: test.user@msubaroda.ac.in, Password: Test@123, Name: Test User, Phone: 9876543210	User created, success message, DB updated (InsertUser)	Pass
UI-FO RM-02	Admin creates user with invalid email	Submit Create User form with invalid email	Email: test@invalid.com, Password: Test@123, Name: Test User	Error: "Invalid email format", form not submitted	Pass
UI-FO RM-03	Admin submits Create User with missing fields	Submit Create User form with empty Email	Email: empty, Password: Test@123, Name: Test User	Error: "Email is required", form not submitted	Pass
UI-FO RM-04	TTM creates timetable entry with valid inputs	Login as TTM, submit Create Timetable Entry form	AcademicClassId: 1, DivisionId: 1, Day: Monday, Time: 10:00-11:00	Entry created, success message, DB updated (InsertTimetableEntry)	Pass
UI-FO RM-05	TTM creates timetable entry with missing fields	Submit Create Timetable Entry with empty AcademicClassId	AcademicClassId: empty, DivisionId: 1, Day: Monday	Error: "AcademicClass is required", form not submitted	Pass

UI-FO RM-06	TTM creates timetable entry with conflicting time	Submit Create Timetable Entry with overlapping TimeSlot	AcademicClassId: 1, DivisionId: 1, Day: Monday, Time: 10:00-11:00 (already exists)	Error: "TimeSlot conflict", form not submitted	Pass
----------------	---	---	--	--	------

SCREENSHOTS

1: Create User Form



Invalid email format.

Create New User

TEST 3

test.com

.....

1234567892

Faculty of Technology and E...

Computer Science and Engi...

RESET

CREATE USER

Create New User

TEST 3

Please fill out this field.

.....

123

Faculty of Technology and E...

Computer Science and Engi...

RESET

CREATE USER

1

Dr. Rahul Sharma

rahul.sharma@msubaroda.ac.in

Faculty of Technology and

Comput Science

Timetable Management System | Testing Report | Prepared by Bharg Goswami 55

2.Create Timetable Entry Form (UI-FORM-04,UI-FORM-05,UI-FORM-06)

localhost:5173/ttm-homepage/timetable-management

Add Timetable

Faculty

Faculty of Technology and Engineering

Department

Computer Science and Engineering

Select Academic Year

☒ 2024-25

Select Program

☒ Master of Computer Applications

Select Class

☒ MCA - Semester 1

Select Division

☒ A

Select Subject

☒ Soft Skills Development

Select Batch

☒ Batch 1

Select Day

☒ Saturday

Select TimeSlot

☒ Slot 5: 03:30:00-04:30:00

Select Location

☒ Room A102

Select Faculty

Faculty of Technology and Engineer...

Select Department

Computer Science and Engineering

Select Teacher

☒ Dr. Anil Patel

☒ Fields with checked checkboxes will keep their selected values when you submit or cancel. Unchecked fields will be cleared.

CANCEL

SUBMIT

Minimum SS for Testing Vite + React

ge/timetable-management

Timetable created successfully!

table

Search Class

Search Teacher

Search Location

4.4 Timetable Management UI Testing

Test Case Table

TC ID	Scenario	Test Description	Input/Case	Expected Output	Status
UI-TT M-01	TTM creates timetable entry	Same as UI-FORM-04, verify DB and UI	AcademicClassId: 1, DivisionId: 1, Day: Monday, Time: 11:00-12:00	Entry appears in timetable view, DB updated	Pass
UI-TT M-02	TTM updates timetable entry	Login as TTM, edit existing entry	Change Time from 11:00-12:00 to 12:00-13:00 for TimeTableId: 1	Entry updated, success message, DB updated (UpdateTimetableEntry)	Pass
UI-TT M-03	TTM deletes timetable entry	Delete existing entry	TimeTableId: 1	Entry removed, success message, DB updated (DeleteTimetableEntry)	Pass
UI-TT M-04	Student views timetable	Login as Student, view timetable	User: amit.prajapati@msubaroda.ac.in, DivisionId: 1	Timetable shows entries for DivisionId=1, no edit options	Pass
UI-TT M-05	TTM views timetable by AcademicClass	Login as TTM, filter timetable by AcademicClass	AcademicClassId: 1	Timetable shows entries for AcademicClassId=1	Pass
UI-TT M-06	Admin Deassigns role	Login as Admin, Deassign role to user	UserId: 21, RoleId: 4 (TTM)	Role Deassigned, success message, DB updated (AssignRole)	Pass

SCREENSHOTS :

1. Timetable View (UI-TTM-01, UI-TTM-02, UI-TTM-03, UI-TTM-05)

Manage Timetable ADD TIMETABLE

CLEAR FILTERS Search Class Search Teacher Search Location DOWNLOAD ENTRIES

Sr. No	Academic Year	Faculty	Department	Program	Class	Division	Batch	Subject	Day	Time Slot	Teacher	Location	Edit	Delete
6	2024-25	Faculty of Technology and Engineering	Computer Science and Engineering	Master of Computer Applications	MCA - Semester 1	B	Batch 2	Database Management Systems	Thursday	Slot 6	Prof. Neha Verma	Room A102		
7	2024-25	Faculty of Technology and Engineering	Computer Science and Engineering	Master of Computer Applications	MCA - Semester 1	B	Batch 1	Data Structures	Saturday	Slot 6	Dr. Anil Patel	Room A101		
8	2024-25	Faculty of Technology and Engineering	Computer Science and Engineering	Master of Computer Applications	MCA - Semester 2	C	Batch 1	Advanced Database Systems	Monday	Slot 6	Prof. Sunita Iyer	Room A102		
9	2024-25	Faculty of Technology and Engineering	Computer Science and Engineering	Master of Computer Applications	MCA - Semester 2	C	Batch 1	Advanced Database Systems	Saturday	Slot 6	Prof. Sunita Iyer	Room A102		

Activate Windows

Update Timetable

Faculty
Faculty of Technology and Engineering

Department
Computer Science and Engineering

Select Academic Year
2024-25

Select Program
Master of Computer Applications

Select Class
MCA - Semester 2

Select Division
C

Select Subject
Advanced Database Systems

Select Batch
Batch 1

Select Day
Saturday

Select TimeSlot
Slot 6: 04:30:00-05:30:00

Select Location
Room A102

Select Faculty
Faculty of Technology and Engineering

Select Department
Computer Science and Engineering

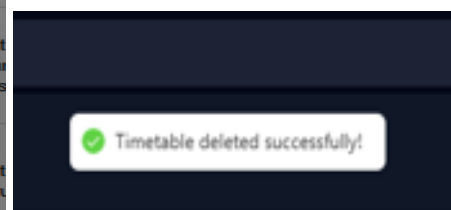
Select Teacher
Prof. Sunita Iyer

CANCEL UPDATE

Delete Timetable Entry

Are you sure you want to delete this timetable entry?

CANCEL DELETE



Timetable

Select Program

Master of Computer Applications

Select Class

Select Location

Room A102

X

Select Teacher

Select Division

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
11:00:00 - 12:00:00	-	-	-	-	-	-
12:00:00 - 01:00:00	-	-	-	-	-	-
01:00:00 - 02:00:00	-	-	-	-	-	-
02:30:00 - 03:30:00	-	-	-	-	-	-
03:30:00 - 04:30:00	-	-	-	-	-	-
04:30:00 - 05:30:00	Advanced Database Systems MCA - Semester 2 Division-C Batch 1 Prof. Sunita Iyer	-	-	Database Management Systems MCA - Semester 1 Division-B Batch 2 Prof. Neha Verma Database Management Systems MCA - Semester 1 Division-B Batch 2 Prof. Neha Verma	-	-

1.SUBJECT

2.CLASS

3.DIVISION

4.BATCH

5.LOCATION

6.TEACHER

DOWNLOAD TIMETABLE

2.Student Timetable View (UI-TTM-04)

Timetable

Full Name: Amit Prajapati

Class: MCA - Semester 2

Division: C

Faculty: Faculty of Technology and Engineering

Department: Computer Science and Engineering

Program: Master of Computer Applications

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
11:00:00 - 12:00:00	-	-	-	-	-	-
12:00:00 - 01:00:00	-	-	-	-	-	-
01:00:00 - 02:00:00	-	-	-	-	-	Artificial Intelligence Batch 1 Room B201 Dr. Anil Patel
02:30:00 - 03:30:00	-	-	-	-	-	-
03:30:00 - 04:30:00	-	-	-	-	-	-
04:30:00 - 05:30:00	Advanced Database Systems Batch 1 Room A102 Prof. Sunita Iyer	-	-	-	-	-

1.SUBJECT

2.BATCH

3.LOCATION

4.TEACHER

DOWNLOAD TIMETABLE

3.Deassign Role Form (UI-TTM-06)

The screenshot shows a web application interface with a table of users and a modal dialog for role deallocation. The table has columns for ID, Name, Faculty, Department, Status, and Role. The modal dialog, titled 'Role Deallocation', contains two dropdown menus: 'Select User' with 'TEST1' selected and 'Select Role' with 'Teacher' selected. At the bottom of the modal are 'CANCEL' and 'DEASSIGN' buttons. The background table shows rows 6 through 10, with row 10 containing 'TEST1' as the user.

ID	Name	Faculty	Department	Status	Role	Assign Role	Deassign Role
6	Dr. Kamlesh Vaishnav	Faculty of Technology and Engineering	Computer Science and Engineering	Active	No Role Assigned	ASSIGN ROLE	DEASSIGN ROLE
7	Aak					ASSIGN ROLE	DEASSIGN ROLE
8	Bha					ASSIGN ROLE	DEASSIGN ROLE
9	Vive					ASSIGN ROLE	DEASSIGN ROLE
10	TEST1	Faculty of Technology and Engineering	Computer Science and Engineering	Active	No Role Assigned	ASSIGN ROLE	DEASSIGN ROLE

Role Deallocation

Select User
TEST1

Select Role
Teacher

CANCEL DEASSIGN

5.Functional Testing

Test Case Table

TC ID	Section	Scenario	Test Description	Input/Case	Expected Output	Status
ROLE-01	5.1	Admin creates role	Login as Admin, create new role	RoleName: "Coordinator"	Role created, appears in role list, DB updated (InsertRole)	Pass
ROLE-02	5.1	Admin deletes role	Delete existing role	RoleId: 7 (unassigned)	Role deleted, removed from list, DB updated (DeleteRole)	Pass
ROLE-03	5.1	Admin assigns role to user	Assign role to existing user	UserId: 10, RoleId: 4 (TTM)	Role assigned, UserRole updated, success message (AssignRole)	Pass
ROLE-04	5.1	Admin deassigns role from user	Remove role from user	UserId: 10, RoleId: 4	Role removed, UserRole updated, success message (DeassignRole)	Passed
TTM-01	5.2	TTM creates timetable entry	Login as TTM, create new entry	AcademicClassId: 1, DivisionId: 1, Day: Friday, Time: 11:00-12:00	Entry created, appears in timetable, DB updated (InsertTimetableEntry)	Passed
TTM-02	5.2	TTM updates timetable entry	Update existing entry	TimeTableId: 1, Change Time to 12:00-01:00	Entry updated, timetable reflects change, DB updated (UpdateTimetableEntry)	Passed
TTM-03	5.2	TTM views timetable by AcademicClass	View timetable with filter	AcademicClassId: 1	Timetable shows entries for AcademicClassId=1 (GetTimetableEntries)	PASSED

TTM-04	5.2	TTM creates entry with TimeSlot conflict	Create entry with overlapping time	AcademicClass Id: 1, DivisionId: 1, Day: Monday, Time: 11:00-12:00	Scheduled "TimeSlot" Is Hidden In dropdown menu while creating new Time Table	Pass
USER-01	5.3	Admin creates user	Create new user	Email: test.user@msubaroda.ac.in, Password: Test@123, Name: Test User	User created, appears in user list, DB updated (InsertUser)	Pass
USER-02	5.3	Admin updates user	Update existing user	UserId: 10, Change Name to "Updated User"	User updated, list reflects change, DB updated (UpdateUser)	Pass
USER-03	5.3	Admin deletes user	Delete existing user	UserId: 10	User deleted, removed from list, DB updated (DeleteUser)	Pass
USER-04	5.3	Admin creates user with duplicate email	Create user with existing email	Email: amit.prajapati@msubaroda.ac.in	Error: "Email already in use", user not created	Pass

SCREENSHOTS :

Role: TTM Logout

Timetable

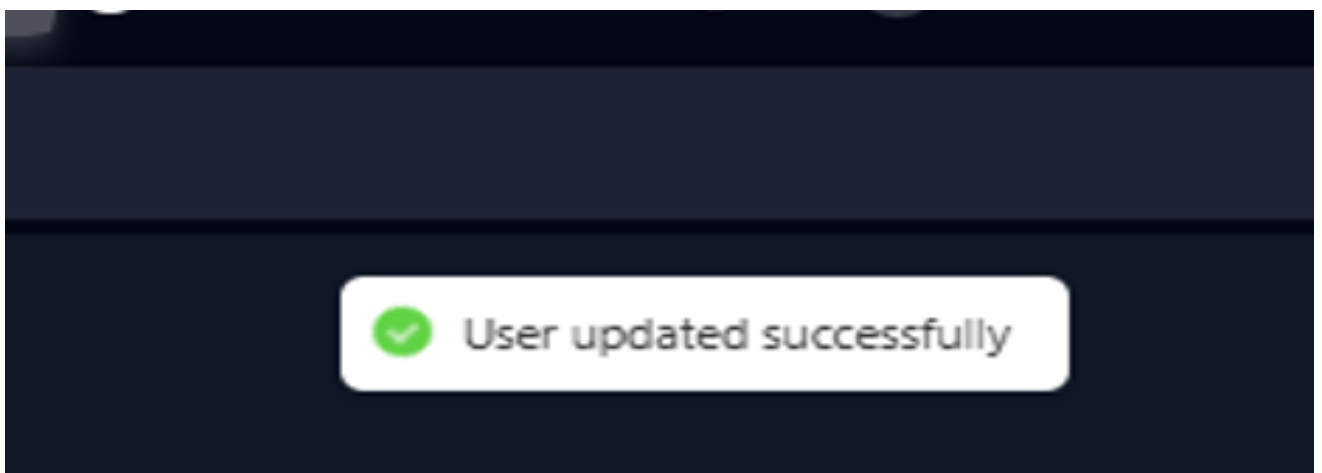
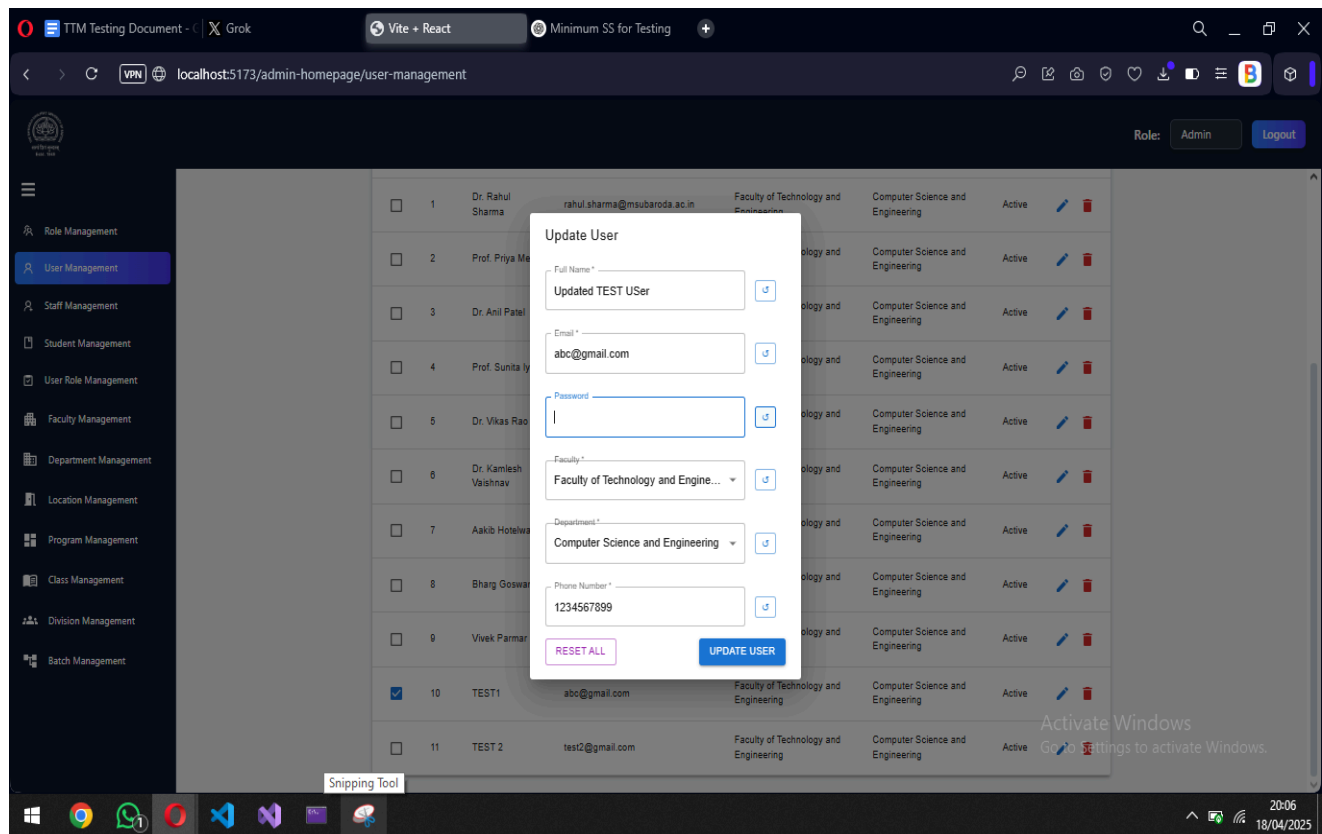
Select Program
Master of Computer Applications

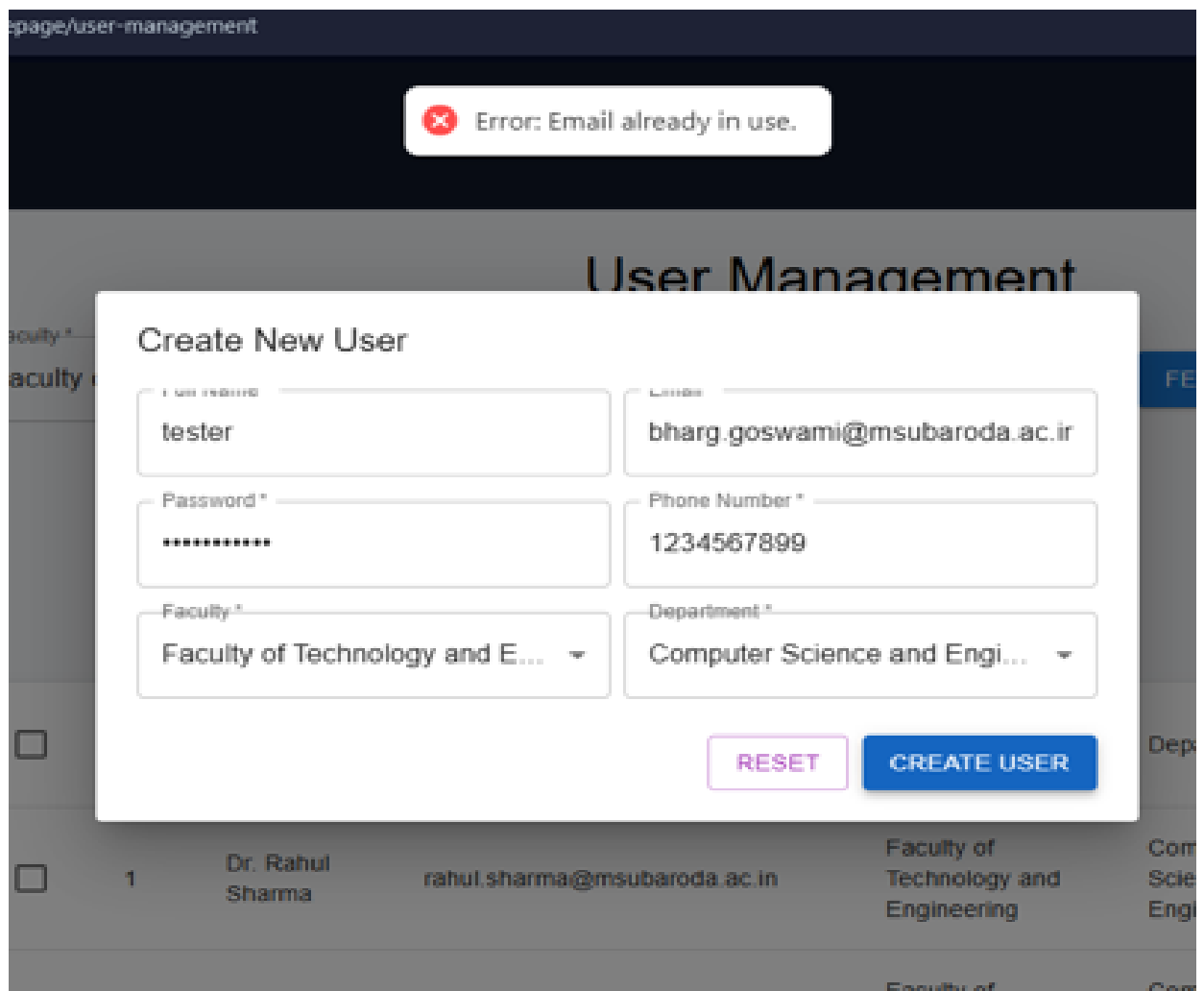
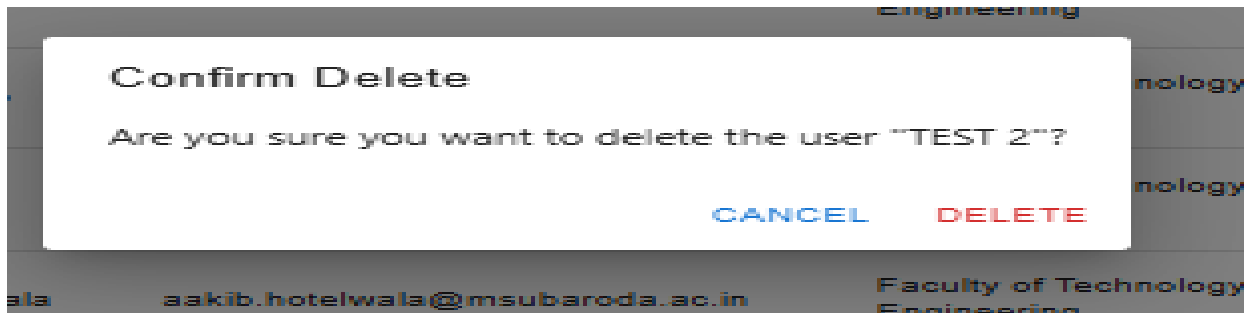
Select Class
Select Division

Select Location
Room A101

Select Teacher

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
11:00:00 - 12:00:00	-	Soft Skills Development MCA - Semester 1 Division-A Batch 1 Dr. Rahul Sharma	-	-	-	-
12:00:00 - 01:00:00	Operating Systems MCA - Semester 1 Division-A Batch 1 Prof. Priya Mehta	-	-	-	Operating Systems MCA - Semester 1 Division-A Batch 1 Dr. Vikas Rao	-
01:00:00 - 02:00:00	-	-	-	-	-	-
02:30:00 - 03:30:00	-	-	-	-	-	-





6. Cross-Browser Compatibility Testing

6.1 Browsers Used

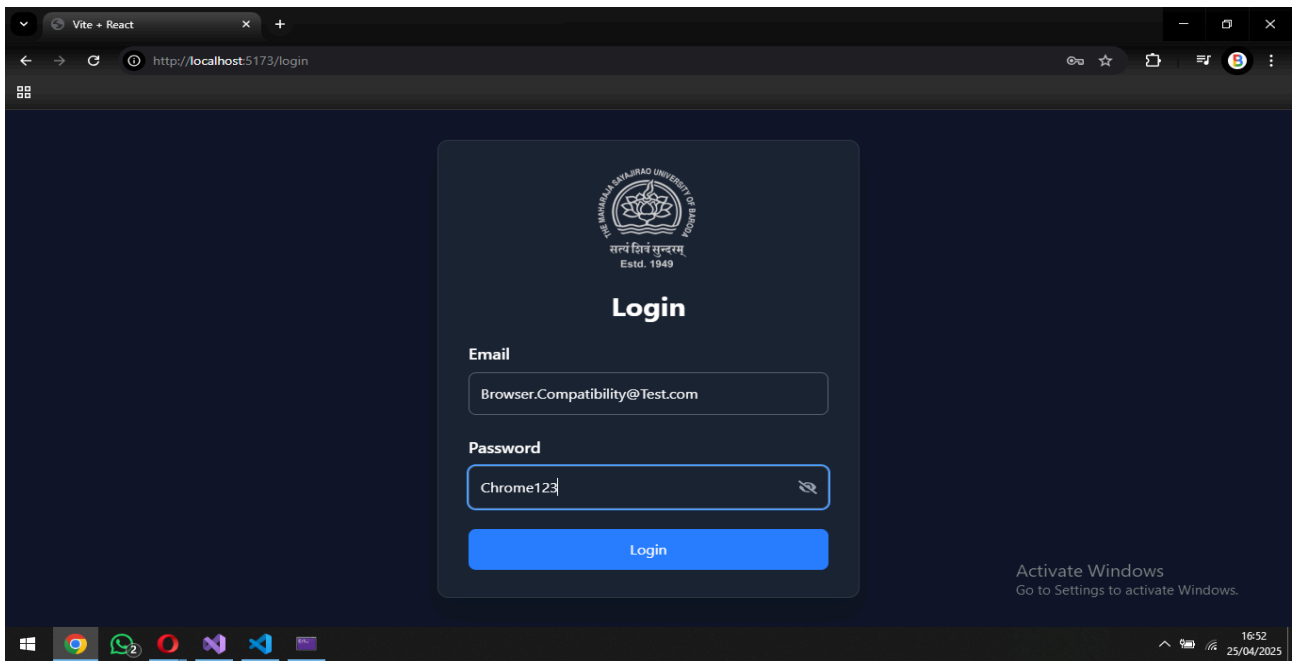
Browser	Version Tested	Status
Google Chrome	1.1+	✓ Compatible
Opera Browser	1.1+	✓ Compatible
Microsoft Edge	1.1+	✓ Compatible

6.2 Cross-Browser Test Cases

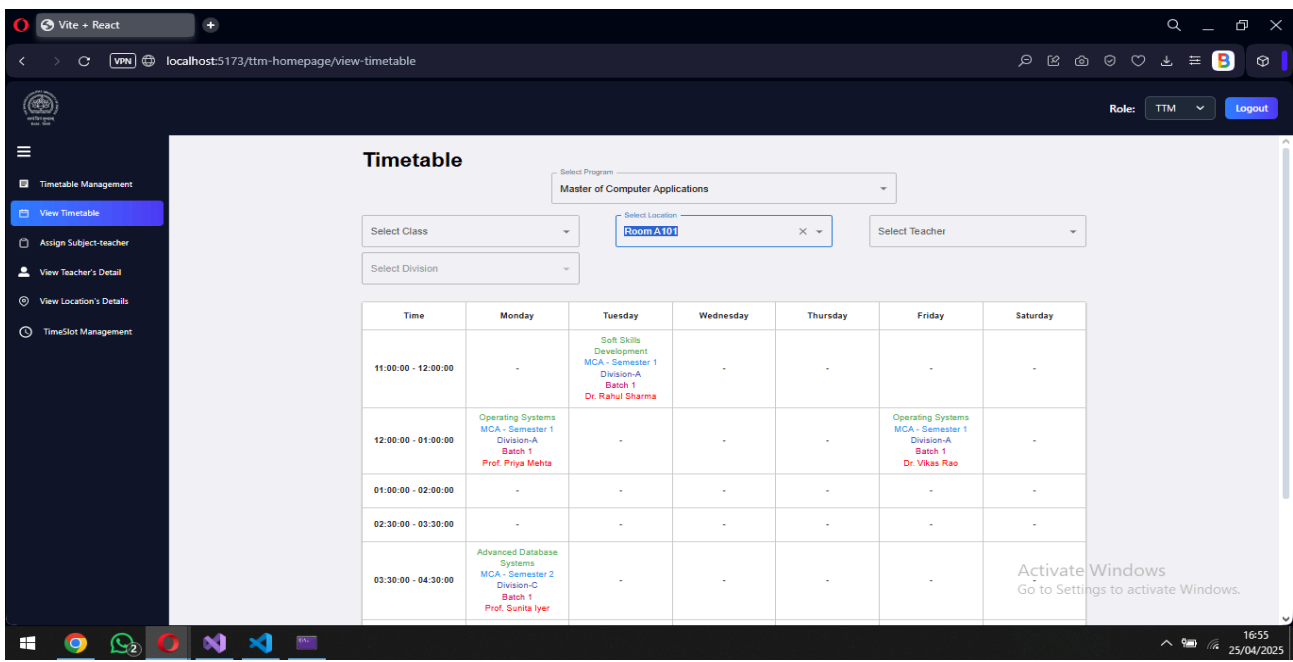
TC ID	Scenario	Browser	Expected Result	Actual Result	Status
CBT-01	Load login page	Chrome	Page loads properly	✓ Same	Pass
CBT-02	Navigate timetable and check alignment	Opera	Elements properly aligned	✓ Same	Pass
CBT-03	Test responsiveness on window resize	Edge	Layout adjusts correctly	✓ Same	Pass

Screenshots :

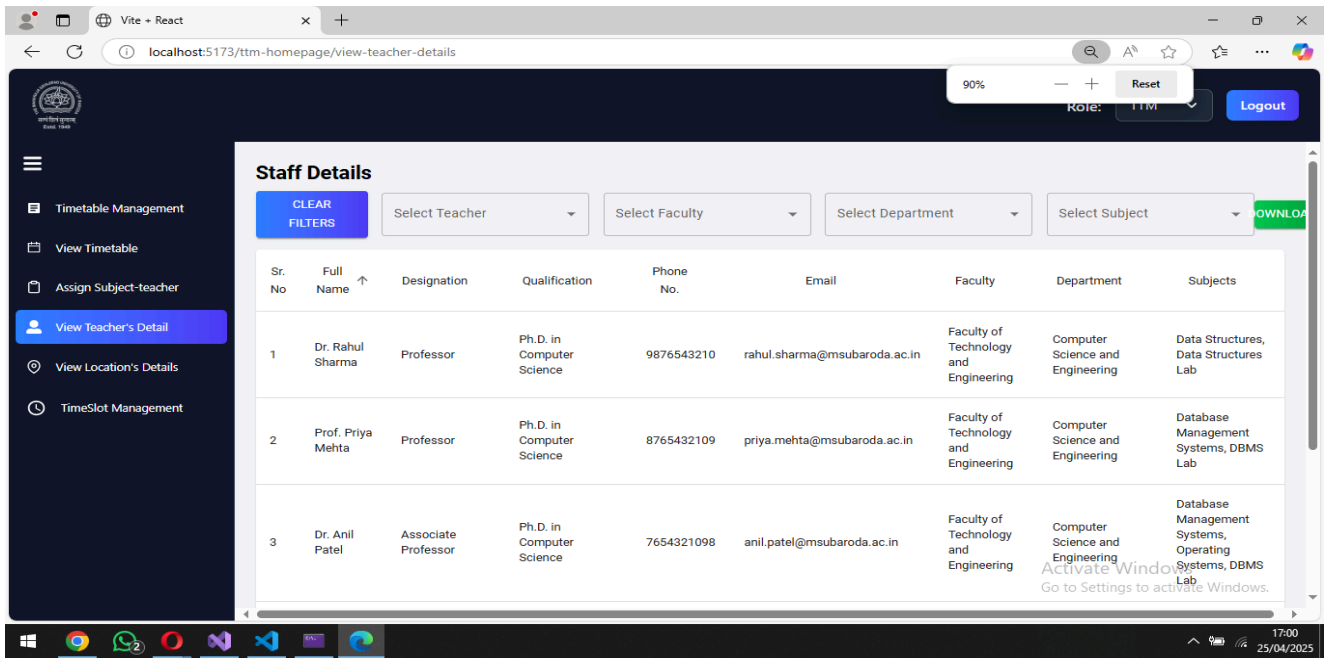
CBT-01 [Google Chrome] LOG IN PAGE



CBT-02 [Opera Browser] TIMETABLE



CBT-03 [Microsoft Edge] Layout Adjustment



7. Bug Tracking & Issues Log

7.1 List of Bugs Found

Bug ID	Test Case	Description	Expected Behavior	Actual Behavior	Status
BUG-01	SP-36	CheckStaffConflict stored procedure output unclear	Return conflict check result (True/False)	As Per Expected Result	 Closed (Resolved)
BUG-02	API-ERR-01	/api/Academic/departments skips validation for missing facultyId	Return 400 Bad Request	Returns empty array [] with 200 OK	 Open
BUG-03	UI-NAV-01 / 08	404 Not Found for /api/Role/roles_permissions and /api/Role/permissions	Return role permissions data	No errors/warnings in Console.	 Closed (Resolved)
BUG-04	UI-NAV-04	Test browser Back/Forward buttons after navigating.	Pages load correctly, state preserved (e.g., filters intact).	Back/Forward button not working properly	 Open

7.2 Severity & Priority of Bugs

Bug ID	Severity	Priority	Notes
BUG-01	Medium	Medium	✓ Fixed in latest SP Version
BUG-02	Low	High	Affects validation and may allow invalid data
BUG-03	High	Medium	✓ Resolved after API controller update
BUG-04	Medium	Low	UI/UX related issue only

7.3 Status Updates

Date	Bug ID	Update Details	Updated By
22-Apr-25	BUG-01	Procedure logic reviewed and resolved	Bharg
23-Apr-25	BUG-02	Issue forwarded to Backend Dev Vivek	Bharg
25-Apr-25	BUG-03	API route added in controller, issue fixed	Vivek
25-Apr-25	BUG-04	UI issue forwarded to Frontend Dev Aakib	Bharg

8. Conclusion & Future Updates

The Timetable Management System (TTM) has been thoroughly tested across all major modules — database, API, UI, and core functionality. Most test cases have passed successfully, and the application behaves as expected in both valid and error conditions. Identified bugs have been logged and most have been resolved.

Project Testing Summary:

- Total Test Cases Executed : **245**
 - Pass Percentage : **~95%**
 - All major CRUD, validation, and role-based operations tested
 - Test evidence includes screenshots and recorded response times
-

Future Updates (Suggested Enhancements):

- Add Permission-Based UI visibility management for better UX
 - Improve error messaging consistency from backend
 - Perform a **Thorough Spell Check And Grammar Review** across the entire system, including UI labels, API messages, and documentation.
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